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Abstract

Using granular supervisory data on deposits, loans, and securities for the largest U.S. banks, we document persistent differences in depositor structures that directly shape asset structures. Retail-depositor oriented banks, for instance, enjoy cheap and stable funding. They thus hold lower-rate, longer-maturity loans and conduct more real estate lending than NBFID-depositor oriented banks. This has aggregate implications. Using exogenous deposit growth around COVID, we show bank depositor orientation shapes their response to deposit growth as well as to interest rate changes. Finally, we augment the paper with a risk measure that captures deviations between asset duration and liability maturity.

JEL classification: G20, G21, G01

Key words: bank funding, lending, depositor flightiness, depositor structure, deposit volatility, financial stability, lending decisions, loan terms

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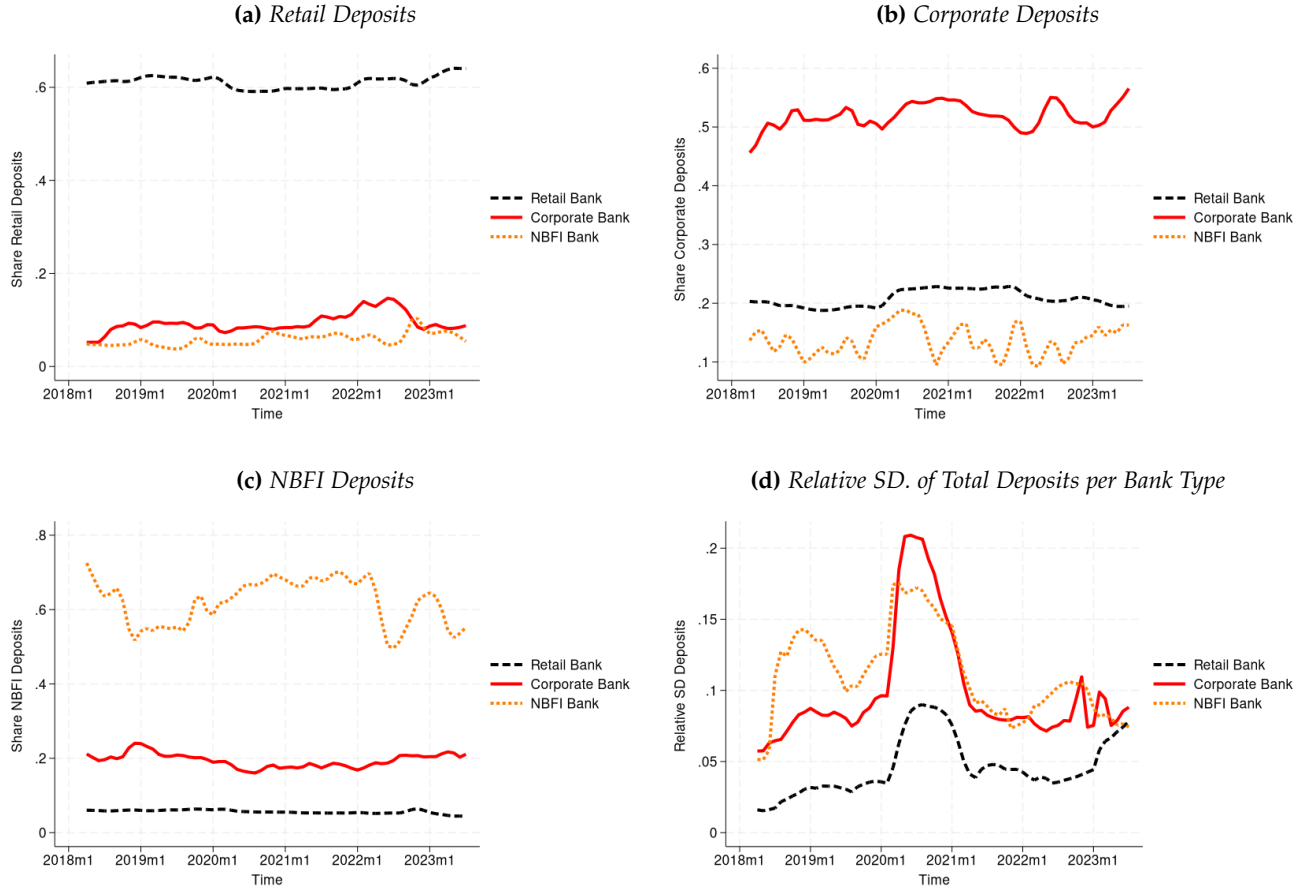
To view the authors' disclosure statements, visit
https://www.newyorkfed.org/research/staff_reports/sr1175.html.

In classical financial intermediation models, banks create liquid demandable deposits while investing the corresponding funds into profitable long-term assets (i.e. maturity transformation). While these demandable deposits provide liquidity insurance for their owners, they expose banks to run risk, which in turn endogenously constrains banks' asset portfolio choices—a fundamental element of models such as in [Diamond and Dybvig \(1983\)](#). However, different depositor types exhibit very different behavior, so the run risk to which banks are *actually* exposed would depend on the types of depositors they serve.¹ Past papers have focused on differences between insured and uninsured depositors, documenting greater deposit flightiness amongst uninsured depositors (e.g. [Benmelech et al. \(2023\)](#)) and highlighting that distressed banks avoid additional uninsured deposits (e.g. [Egan et al. \(2017\)](#)). However, the concept of "uninsured depositors" can be a very rough proxy for depositor counterparty-types, which can behave very differently at different types of banks. Moreover, bank asset duration, which traditional banking wisdom suggests must match depositor flightiness, can be hard to glean from aggregate data. Understanding (i) depositor composition differences across banks and (ii) how depositor behaviors align with asset duration broadly (and loan terms) is of fundamental importance to financial intermediation research and macro-prudential policy. *Which* type of deposits grow at *which* bank – especially following shocks or fiscal stimuli – will determine who can obtain credit at what price.

In this paper, we are the first to comprehensively answer several key questions *together* using granular supervisory data on bank depositors and assets. We document three relatively novel findings of use to fellow researchers and policymakers. First, we show that even large banks can cater to very different depositor types. This is illustrated clearly in [Figure 1](#), which uses monthly depositor filings for the 50 largest US banks. We group banks based on their largest single depositor counterparty as of January of 2020—for most banks that is a depositor counterparty type that represents over 50% of deposits at the bank. Banks differentiate themselves by catering to distinct depositor segments: retail-depositor-oriented banks have almost no NBFI depositors, and vice versa. The differences in depositor structure are a persistent, time invariant feature of each bank's business model. Depositor counterparty type correlates with preferences for depositor product type and insured vs. uninsured deposits. In fact, depositor types subsume traditional insured/uninsured dichotomy, especially when we consider that depositor types behave differently depending on the bank-type in question. On average, retail banks have more stable

¹[Chen et al. \(2022\)](#); [Drechsler et al. \(2023\)](#) find that insured and uninsured depositors generate different run risks. [Blickle et al. \(2025\)](#) show that rate-sensitive depositors increase aggregate run risk. Early work by [Hannan and Berger \(1991\)](#) and [Neumark and Sharpe \(1992\)](#) shows that banks can price deposit contracts differently depending on their depositor base, with franchise or market power playing a clear role—an insight laid out in [Drechsler et al. \(2017\)](#).

Figure 1: Bank Types based on Deposit Structure



Notes: This figure groups banks in our sample into three types based on whether they obtain the plurality (usually more than a half) of their deposits from retail, corporate, or NBFI depositors. For instance, a retail bank obtains the largest share of its deposit funding from retail depositors. We ignore small business, government, bank, and other depositors in this analysis for simplicity. NBFI depositors include brokerage houses, insurance companies, independent mortgage lenders, and any other financial firm that is not itself a bank. Panels (a)–(c) show the share of deposits at the given bank-type held by each depositor type. Panel (d) shows the relative standard deviation (computed as the six month rolling standard deviation of deposits scaled by mean total deposit volume) for each bank type. In all Panels, bank types are defined as of January 2020.

depositor bases and pay lower average deposit rates than NBFI- or corporate-deposit-oriented banks. NBFI- and Corporate-depositor oriented banks face increased depositor outflow pressure in times of higher aggregate rates and tend to aggressively compete for deposits to accommodate their expensive funding. Retail-depositor-oriented banks exhibit show far less depositor flightiness and an insignificant response to rate hikes.

Second, bank funding influences bank asset allocation choices. Retail banks tend to hold more corporate loans than even corporate-depositor oriented banks, invest more in real estate, and hold fewer securities. The longest maturity or lowest rate C&I loans are found at retail-depositor oriented banks. NBFI-oriented banks, by contrast, hold almost exclusively treasury securities, reflecting the volatile nature of their depositor base. We complement these findings with a novel risk measure that tracks the

gap between asset durations and liability maturities. We use recent events to show that growth in this gap can be highly predictive of bank distress.

Third, depositor specialization shapes how banks respond to aggregate funding shocks. Using the quasi-exogenous deposit growth generated by the COVID-era quantitative easing (QE) and stimulus payments, we show that funding shocks do not translate uniformly into credit supply. Retail-depositor-oriented banks invest newly arriving deposits into real estate loans. Corporate banks channel new corporate deposits primarily into C&I lending. NBFIs respond very little to new deposits, preferring to hold cash and reserves. These heterogeneous responses have real consequences: firms borrowing from corporate-deposit-oriented banks that received funding inflows grew their assets and profitability during COVID, while firms borrowing from retail-oriented banks did not experience the same gains in C&I-driven growth.

Taken together, our findings imply that banks are not interchangeable conduits for credit or monetary policy. Asset and liability choices are jointly determined and mutually reinforcing, so the allocation of deposit inflows — such as those generated by QE and COVID relief efforts — depends critically on which banks receive them.

The data we use for this paper comes primarily from 2052a monthly reports by banks above 100 billion in assets, which includes asset and liability details. Most crucially, it breaks deposits down by counterparty-type*insured*maturity*product-category for each bank. We further use Y14 data collected as part of the FRB’s stress testing efforts. We make use of details on securities holdings as well as C&I lending. We round out our analyses with publicly available information taken from call reports and Y9C public filings.

The existing literature shows that a bank’s depositors base influences both its resilience and its ability to lend. Sticky deposits create franchise value that supports long-term lending when rates rise ([Drechsler et al., 2021](#)), and pooling liquidity needs across depositors and credit line users lowers the cost of managing withdrawals and drawdowns ([Kashyap et al., 2002](#)). Consistent with this, banks with more stable funding appear more willing to engage in longer-term, relationship-based lending (e.g., [Gatev and Strahan, 2006](#)). By contrast, uninsured and information-sensitive depositors are more prone to run on signals of asset deterioration (e.g., [Iyer and Puri, 2012](#); [Iyer et al., 2016](#); [Martin et al., 2013](#); [Blickle et al., 2024](#); [Carletti et al., 2023](#); [Jiang et al., 2024](#)), and policy uncertainty can further amplify withdrawals among certain depositor segments ([Arvanis et al., 2022](#)). Recent work using granular deposit-level data, almost as granular as ours, reinforces the idea that depositor composition matters: larger depositors

are more flighty (Cipriani and Eisenbach, 2023); rapid growth from large individual depositors can allow banks to downsize their branch networks but increases outflow risk in stress periods (Benmelech et al., 2023); and banks appear to sort into distinct business models—high-rate/low-branch versus low-rate/high-branch—to attract different types of depositors (Kundu and Zhang, 2024).² Similar to our paper, Chen et al. (2024) use public data to show that uninsured depositors are particularly sensitive to negative return shocks when the bank is invested in more illiquid assets. However, all this work typically treats depositor bases in broad aggregates—e.g., insured vs. uninsured, wholesale vs. retail—without examining the internal heterogeneity of depositor segments or how banks strategically specialize in serving them. Regulatory frameworks such as the Liquidity Coverage Ratio (LCR) acknowledge that deposit stability varies, but they do so only coarsely and without linking depositor composition to bank business models, portfolio choices, or lending outcomes. Ours is the first paper to dig into granular deposit type differences, showcase their volatility and what it implies for the overall funding stability of the banks in question.

At the same time, complementary literature shows that banks specialize on the asset side by focusing on specific borrower types, industries, or geographies. It is well documented now that specialization and deepened relationships with customers reduces information production costs (e.g., Berger and Udell, 1995; Petersen and Rajan, 1995; Degryse and Ongena, 2005; Paravisini et al., 2023; Blickle et al., 2025, 2024). Some recent papers have begun to bring liabilities into this picture by emphasizing how funding models, deposit stickiness, and rate sensitivity affect lending behavior and monetary policy transmission (e.g., Jiang et al., 2023; Egan et al., 2023; Drechsler et al., 2017; English et al., 2018). Where the first strand of the literature takes funding almost entirely as given, the second strand of studies are unable to treat depositors with the granularity we are, usually distinguishing only between insured and uninsured.

Our paper brings these strands together by taking a detailed view of both sides of the balance sheet. We show that banks specialize not only in how they lend but also in whose deposits they take. Nuanced differences in the depositor-base are tightly linked to the type of assets a bank buys and even the loan terms of individual deals. Using highly granular supervisory data, we document that depositor

²Recent work has also begun on the influence of fintechs on banks: Koont et al. (2024) divide banks into categories based on whether they offer an app or brokerage accounts. Banks with digital platforms experience faster deposit outflows when rates change. Similarly, Erel et al. (2023) compare FinTech online banks with traditional banks. They find that the former increase deposit rates significantly more than traditional banks following Federal funds rate increases. Lu et al. (2025) shows that payment conveniences increases depositor flightiness and reduces account balances. Narayanan et al. (2025) show that branch closures have been concentrated in markets with technologically savvy investors.

composition is a persistent and defining feature of bank business models—one that co-evolves with lending specialization rather than simply constraining it. This fills a key gap: prior work recognizes that funding stability and depositor flightiness matter, but it has not shown how variation in who provides the deposits translates into systematic differences in how banks lend. Nor has the literature been able to show that this has real aggregate implications for the real economy. How deposit growth is intermediated depends on how deposits are growing at what bank. This perspective implies that banks are not interchangeable conduits for credit or monetary policy. Because asset and liability choices are jointly determined and mutually reinforcing, the allocation of deposits (think QE and relief efforts during COVID that affected funding structures ([Greenwood et al., 2016](#); [Rodda et al., 2022](#)) .

1 Data

1.1 C&I Lending – Y14-Q, H

Our loan-level C&I lending data is taken from the Y-14, schedule H. This data is maintained by the Federal Reserve for stress testing purposes. It tracks all C&I loans, originated or held by stress tested banks, over 1 million USD in size. It includes term loans and revolving credit lines. For each facility, the data includes borrower identifiers, industry codes, commitment and utilization amounts, interest-rate terms, collateral/guarantee data, origination and maturity dates, loan purpose information, and performance status (since loans tracked over time). Importantly, the data also contains details on the bank’s internal loan rating, which is standardized across banks for the stress testing efforts. Data is collected at a quarterly frequency. Importantly, unlike commercial data sets, we can observe lending to both public and private businesses. After merging our datasets (see below) we observe a slightly unbalanced panel of 25 institutions. These 25 banks are the largest in the US C&I lending space and represent the vast majority (>65%) of US corporate lending.

We use our loan-level data in two ways. First, we collapse our data to the bank-quarter level. This allows us to construct asset-side composition measures by loan type, by rating bucket, and by maturity. Second, we use our loan-level data directly and look at loan pricing (interest rate), loan size and loan maturity. The more granular approach allows us to account for more characteristics of the loan.

1.2 Securities Holding Data – Y14-Q, B

Y-14 data includes information on the securities held by individual banks. Schedule B includes details that allow us to identify the specific security purchased by the bank in a given quarter. The data tracks purchase price and – where relevant – the current market value at quarter's end. For simplicity, we group the data into categories such as US treasuries, Mortgage backed Securities (MBS), (which together account for over 80% of securities held by banks in our data), as well as corporate bonds, and CLO investments. For most analyses, detailed below, we further collapse our data to the bank-quarter-security-type level. This is because additional details regarding the securities themselves are sparse or not useful for our purposes.

1.3 Deposit Data – 2052a, Liabilities

We complement our asset data with high-frequency funding information from FR 2052a (also known as the Complex Institution Liquidity Monitoring Report). The 2052a includes information on a bank's contractual liabilities. We focus on deposits data, for which detailed granular breakdowns are available. While we cannot observe individual depositors, the data is broken into several useful categories. Specifically, we observe deposits at the maturity * insured/uninsured * product type * counter-party level. Maturities are mapped into standardized time buckets (e.g., open, 2–7 days, 8–30 days, etc.). We typically focus on "open" (i.e. demandable) vs time deposits. Counter-parties can be broadly grouped into banks, public sector entities, small businesses, NBFI's (which include funds and non-bank lenders), corporate depositors, and retail depositors.

While the largest systemically relevant banks report daily data, others report at lower frequencies. For our purposes, we primarily use monthly data, aggregating information from more frequent reporters where necessary. This sample is more consistent (daily reporters can change their reporting frequencies) and shows sufficient variation for our purposes. We can use deposit data from 2018q1 to around 2024q1. Some extensions can be performed with earlier data, though changes in reporting structures complicate certain analyses.

We separately collapse our data to the bank*counter-party*month, bank*insured*month, bank*maturity*month level before calculating the key metrics (such as variance) which we use below. Subsequently, we collapse our data further to the bank*quarter*type level for merging with quarterly data sets.

1.4 Aggregate Bank Data – Y-9C

We merge the data, discussed above, with confidential Y-9C filings for bank holding companies. This data includes details on bank size, on risk total weighted assets, and on real estate lending, for which we currently have no more granular a breakdown. The confidential version of the Y-9C is slightly more detailed than the public version, which is useful for the capital measures and risk weights we analyze below.

Our merged sample contains 25 banks (after dropping non-domestic US banks or inconsistent filers of the 2052a). Total deposits of banks in our clean, merged sample account for 11.3 trillion USD of the approximately 17 trillion USD in deposits held by all banks at the end of 2022. For analyses that use deposit data that is not merged with Y-14 or Y-9c data, our deposit data accounts for 13 trillion USD of deposits in 2022. When showing

We show key summary statistics of variables – for our fully merged sample – in Table 1. We can see that there exists a large heterogeneity both in how banks in our sample invest funds and in how (i.e., from whom) they obtain deposits. Retail deposits make up around half of all deposits in our sample with significant heterogeneity across banks. just over 40% of deposits are insured – after all, there are several retail deposit accounts that are not insured. Interestingly, the vast majority of funds in our sample are demandable. The relative standard deviation (i.e. the 12 month standard deviation / 12-month mean deposit base) of retail deposits is much lower than NBFI deposits.

Around a third of lending in our sample is traditional C&I lending. The average size of our logged loans is 8.8. The average interest rate for loans in our sample period is 4.4%, a figure pulled up by recently higher rates. Around 25% of loans use no form of collateral while 40% can be classified as term loans.

2 Methodology

2.1 Bank-Level Analyses

We begin with simple bank-level analyses that test the relationships between a bank’s funding structures and asset structures. Specifically, we relate the share of a bank’s balance sheet devoted to (i) C&I lending (ii) Securities held (iii) Cash or Reserves held and (iv) real estate lending to the share of a bank’s deposits that are either corporate, NBFI, retail, insured, or demandable deposits. For bank b in quarter t we run

Table 1: Summary Statistics of Key Variables

	N	Mean	SD	25-pct	75-pct
Aggregated Assets					
C&I lending share	500	0.31	0.16	0.18	0.43
Cash/reserves share	500	0.12	0.09	0.06	0.15
Real estate share	500	0.16	0.11	0.07	0.25
Securities share	500	0.20	0.08	0.15	0.24
Aggregated Liabilities					
Retail deposit share	500	0.49	0.24	0.41	0.59
Corp. deposit share	500	0.20	0.10	0.13	0.29
NBFI deposit share	500	0.16	0.24	0.03	0.18
Insured deposit share	500	0.42	0.21	0.32	0.53
Demandable/ST share	500	0.89	0.10	0.86	0.96
σ/μ total deposits	500	0.05	0.03	0.02	0.06
σ/μ retail deposits	500	0.05	0.10	0.02	0.06
σ/μ corp. deposits	500	0.08	0.07	0.04	0.11
σ/μ NBFI deposits	500	0.10	0.09	0.05	0.12
Loan-Level					
Interest rate	3,339,792	4.36	2.99	2.78	5.50
Loan maturity remaining	3,339,792	13.9	12.6	4	19
Log size	3,339,792	8.82	1.48	7.57	9.90
Unsecured loan	3,339,792	0.25	0.43	0	0
Term loan	3,339,792	0.40	0.49	0	1

Note: This table shows summary statistics for banks and loans in our sample. We collapse bank-level variables to the quarter-bank level (even if they are computed at a greater frequency). Lending shares are the portion of a bank's total assets devoted to a certain type of asset-class. Includes all commitments and outstanding loans to corporates, as measured in the Y-14 H data. Cash and reserves are combined and taken from Y-9C disclosures. Similarly, real estate lending represents all real estate (commercial and residential) as detailed in Y-9C data. Securities are the current market value of treasuries, MBS, corporate bonds, CLO investments or all other commercial paper held by the bank. On the liabilities side, we compute the share of deposits held by various counterparties (leaving out bank, small business, public entities, and other small groups for convenience), as well as by insured vs uninsured and by demandable (open or maturity of 1 day) vs time deposits. Finally, the relative standard deviation of total deposits is calculated using a 12 month rolling window for the sd and the mean deposit level and collapsed to the quarterly level using the final observation in each quarter. For loan-level data, we count each bank-loan-time combination. The interest rate is the un-adjusted cost of the loan, measured in percent. Maturity is measured in quarters remaining. Unsecured is a dummy that takes the value of 1 if the loan is not secured with any type of collateral.

separate regressions of the form:

$$Y_{bt} = \eta DepositShare_{bt} + \delta_b + \epsilon_{bt} \quad (1)$$

We are looking to identify η , which tracks the degree to which a bank's deposit makeup affects its asset makeup. As discussed above, one might conjecture that deposits, which are traditionally more stable (such as retail deposits), may allow a bank to invest in more long-term projects. The run risk – and consequently the costly liquidation risk – is much lower. We can account for bank fixed effects that remove any time-invariant bank characteristics. In the face of bank fixed effects, we are identifying the balance-sheet effect of a relative change in a bank's funding structure. In order to cleanly identify our effect of interest, we can further extend the analysis to include controls for bank funding characteristics such as its capital ratio, its risk weighted assets to total assets, and its size.

We can extend the idea behind the above analysis and integrate deposit changes as a key variable. After all, we are interested in seeing not only whether different bank-types have different asset structures but also whether different bank-types respond differently to deposit changes. Our regression of interest, using the same dependent variables outlined above, becomes:

$$Y_{bt} = \eta DepositShare_{bt} + \zeta DepositChange_{bt} + \beta DepositChange_{bt} * DepositShare_{bt} + \delta_b + \epsilon_{bt} \quad (2)$$

Where our variable of interest is now β , which shows how banks with different depositor specializations may respond to changes in various types of deposits. We run separate regressions for changes in retail, corporate, NBF, or insured deposits. We can again account for bank fixed effects and also its capital ratio, its risk weighted assets to total assets, and its size in extensions. However, given the small number of observations available to us at the bank*quarter level, we cannot saturate our regression with a large number of controls or fixed effects. To this end, we also use analyses at the loan/security level.

2.2 Loan-Level Analyses

At the loan-level, we relate a loan's (i) interest rate, (ii) the maturity on the loan, and (iii) the size of the loan to the balance sheet composition of the bank. As above, we are particularly interested in the share of the deposits a bank has obtained from NBF depositors, corporate depositors, retail depositors, or insured depositors. However, unlike in the regressions above, we are able to account for a host of controls and fixed effects. First, we are able to account for loan terms that are not the dependent

variable, acknowledging that all terms are determined simultaneously during a negotiation. Second, we can control for the collateral pledged as well as industry*time, loan purpose, loan-type, and risk rating fixed effects.

In saturated extensions, we can also include bank fixed effects. This allows us to make inferences about the effect that a relative change in a bank’s funding structure may have on C&I loan terms. The regressions would take the following form for loan i , to firm operating in 4-digit NAICS industry j , at time t , made by bank b :

$$Y_{ibjt} = \eta DepositShare_{bt} + \alpha Collateral_i + \alpha Terms_{ij} + \delta_{jt} + \delta_b + \alpha Purpose/Type_j + \epsilon_{ibjt} \quad (3)$$

We again may expect a bank with more stable funding (such as banks with a higher share of retail deposits or a larger share of insured deposits) to offer longer maturity loans – all else equal. Similarly one may expect banks with funding that is less responsive to aggregate rate changes – and therefore less costly – to offer more generous loan rates (or larger loans for a given rate). We can run the regressions for all loan-observations, including re-negotiations that may occur during times when a bank’s funding structure undergoes changes, as well as new observations only.

As above, we are also interested in teasing out the degree to which a bank may respond to changes in its deposit base. We can again interact our variables of interest (the share of a bank’s deposit from a specific source) with changes in various deposit types. Given the greater number of observations in our loan-level data, we can specifically focus on the COVID-era, which saw unprecedented deposit growth. Since much of this deposit growth was unsolicited, it gives us a view into how different types of banks allocate exogenous deposits.

3 Depositor Concentration and Deposit Stability

In this section we explore the fact that banks specialize in catering to certain types of depositors. We show that this has implications for how stable bank depositor structures are and how

3.1 Documenting Bank Depositor Type Preferences

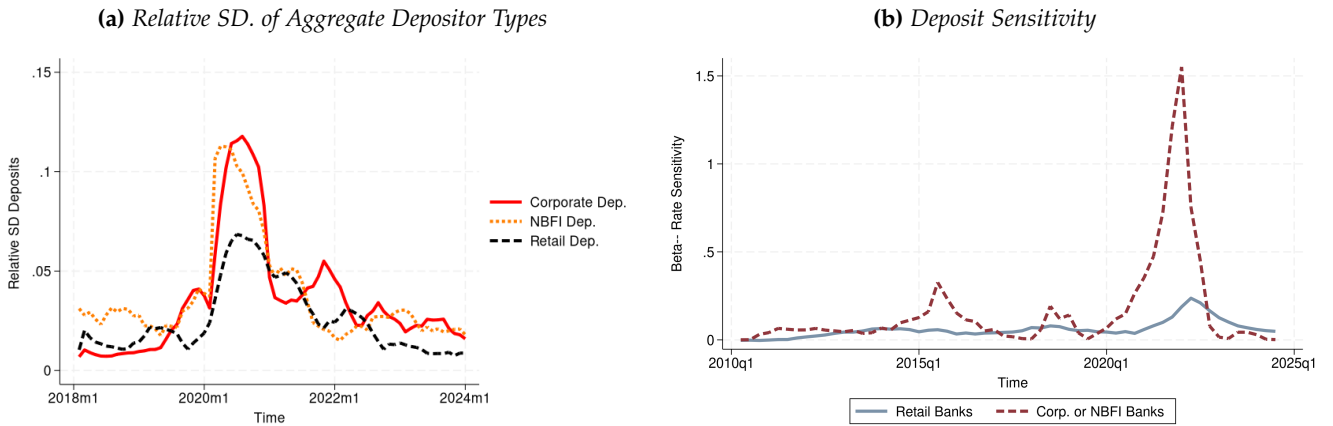
As discussed in Figure 1 above, banks show a clear preference for certain types of depositor counterparties, and this specialization in depositor types can have implications for the stability of a bank’s

deposit base. After all, different types of deposits have – on average – different degrees of volatility.

We plot the relative variability of deposits in Figure 2 panel (a). We define relative variability as the 12-month standard deviation in deposit type over the 12-month mean deposit volume. Panel (a) shows that retail deposits enjoy the lowest variability on average. This held even during 2020, when aggregate volatility rose substantially. This can be thought of as an alternate version of panel (d) from 1. Ostensibly it documents *why* banks that cater to different types of depositors have different aggregate deposit volatility.

Panel (b) takes a somewhat different approach. Here we plot the sensitivity of deposit flows to rate changes. As in Drechsler et al. (2017), we instrument deposit rates with fixed costs at banks. Given limited instrumentation power in smaller samples, we combine NBFI and Corporate deposits. NBFI and Corporate deposits are typically amongst the most volatile in ordinary times, as these entities derive little "convenience" from being depositors. Our approach here follows Blickle et al. (2025). We can see that NBFI and corporate depositors are far more rate sensitive on average, with this rate sensitivity spiking during COVID-QE (something we touch on again below)³.

Figure 2: Depositor Types behave Differently



Notes: Panel (a) depicts the relative standard deviation (12-month rolling window standard deviation over 12-month mean deposit level) for different depositor types. Panel (b) shows the responsiveness of depositors to rate changes. We instrument interest rates paid by banks with bank fixed costs and regress deposit volumes of different depositor counter-parties at banks on rates. We combine NBFI and Corporate deposits given weak instrumentation power in smaller samples.

In Appendix Figure A.2, we show the maximum draw-down per depositor type in any given quarter across all banks in the sample. We smooth the lines for ease of viewing. The maximum flight of retail depositors from any bank in our sample is always smaller than that of corporate or NBFI depositors. In fact, corporate depositors are liable to engage in the largest single month-to-month outflow.

³In this analysis we make use of Call -report data on interest paid by banks. We cannot differentiate the interest paid for different types of deposits. This figure should be seen as indicative.

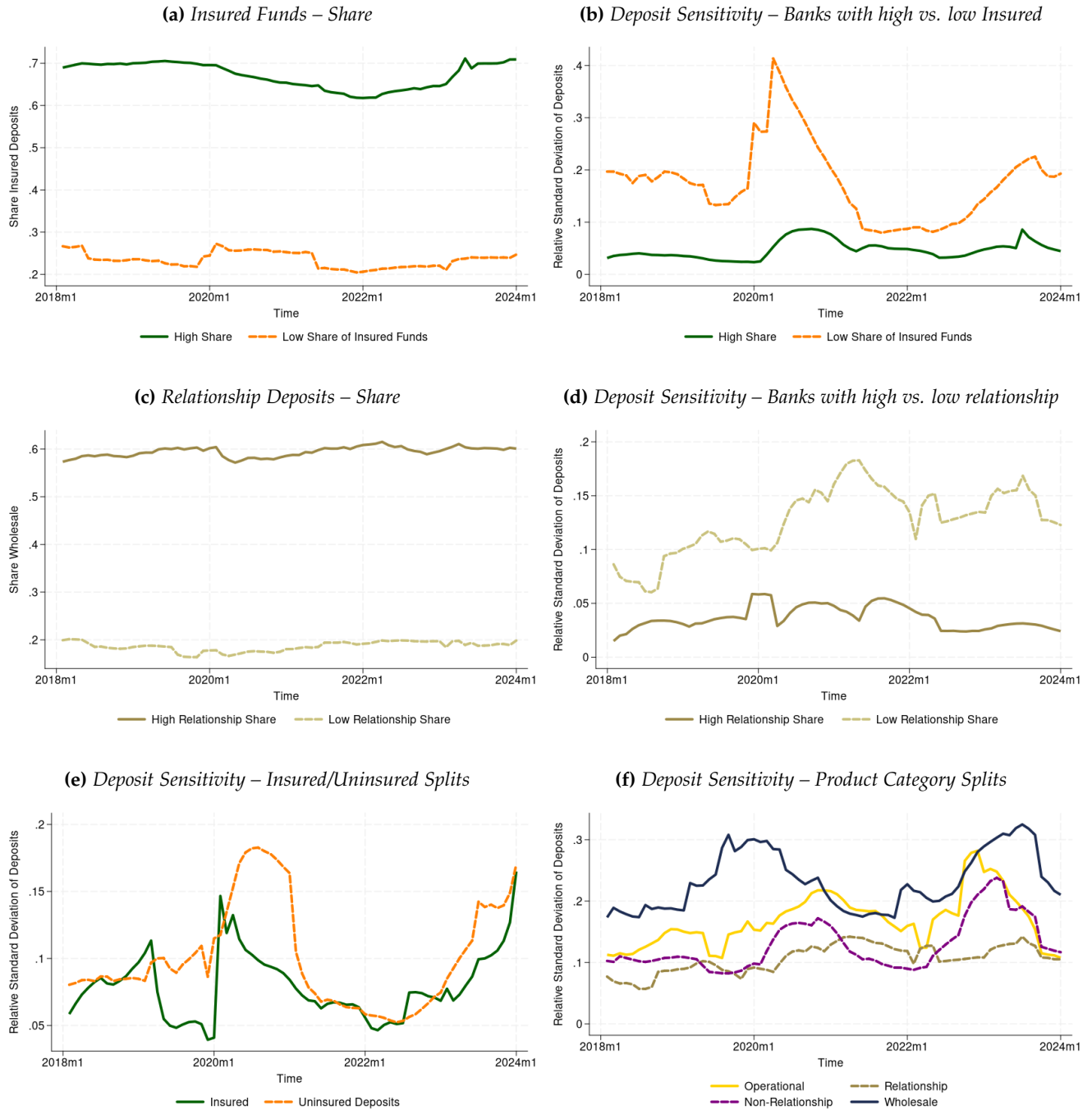
We see that the relationships discussed above are not driven by individual outliers. Instead, in Appendix Figure A.3, we relate the average 12-month standard deviation of deposits in any given bank-quarter observation to the share of the balance sheet comprised of either retail- (panel (a)), corporate- (panel (b)) or nbfi-depositors (panel (c)). The relationship is almost linear: the greater the share of corporate- or nbfi-depositors the greater the average variability of a bank's deposits. Conversely, the greater the share of retail depositors, the lower the average deposit variability. This holds in levels as well as in changes, as can be seen in Appendix Figure A.4, where an increase in volatile depositor types increases the bank's deposit volatility.

We make use of either aggregate deposit volatility or splits based on depositor counterparty type in much of the body of this paper. Splits by depositor type are (i) novel in the literature (ii) intuitive and (iii) most importantly, explain perhaps the largest share of aggregate deposit variability in banks. We document this in Appendix Table A.1, wherein we show that if we include depositor counterparty types, then the deposit products a bank offers and the share of deposits that are insured become less important in explaining aggregate deposit volatility. This is because depositor types *explains* whether deposits are likely to be insured or what products are likely to be used. Retail depositors, after all, are unlikely to use uninsured wholesale accounts. We document this illustratively in Appendix Figure A.5, where we see that depositor types are strongly correlated with insurance coverage and product categories. However, for the purpose of completeness, we include Figure 3 where we show bank preferences for different categories of deposit products and insured vs uninsured deposits.

As we can see from Panel (a), where we split banks into quartiles based on the share of their depositors that are insured, certain banks make use of a large share of insured deposits while some banks do not. This has implications for the aggregate volatility of bank deposits, wherein banks with a greater share of uninsured deposits have a higher (and more volatile) relative standard deviation of deposits (see panel (b)). Similarly, relationship deposits are more stable than non-relationship deposits. Some banks make use of very few while some banks make use of overwhelmingly relationship deposits. Finally, we show the relative standard deviation of insured and un-insured deposits (e) and the relative standard deviation of different product types (which include relationship, wholesale, operational and non-operational deposits).

What can be inferred from the chart above is that banks specialized in a certain type of deposit have lower average deposit volatility than could be expected based solely off of the deposit volatility of the deposit type. There appears to be an additional stability benefit conferred by specializing in a certain

Figure 3: Depositor Types –Alternate Splits



Notes: Panel (a) depicts the share of deposits that are insured at banks in either the top or bottom quartile of banks with insured deposit shares in January of 2020. Panel (b) depicts the relative standard deviation of total deposits for the same two categories of banks. Panel (c) depicts the share of relationship deposits at banks with the highest and lowest quintiles of relationship deposit shares in 2020. Panel (d) depicts the relative standard deviation of total deposits for the same two categories of banks. Panels (e) and (f) depicts the relative standard deviation of deposits in each of the depicted deposit categories.

type of deposit.

3.2 Benefits to Specializing in Depositor Types

It seems then, that to a certain extent specialization in a certain type of depositor reduces the flightiness of that depositor type. Some of this may result from the same logic that drove results in [Blickle et al. \(2024\)](#). A bank focused on a certain depositor type may manage that depositor better, know how to offer terms that keep those depositors, or even just provide the infrastructure necessary to facilitate the depositor, etc. We can see this dynamic at play in [Figure 4](#), where we relate the share of deposits a bank receives from a certain type of depositor to the volatility of that depositor type. The more specialized a bank is in corporate deposits, for instance, the smaller the volatility of its corporate depositor base.

Part of this relationship is likely mechanical. A very small base of a certain depositor type will naturally mean that even small variations in these deposits translate to large relative standard deviations. This is part of the reason that the largest effect appears driven by banks with very few of a certain type of deposit. However, the relationship holds, if more weakly, at higher thresholds too. We can show this in [Appendix Table A.2](#) where we include controls and even bank fixed effects. An increase in the share of deposits from a single counterparty reduces the volatility of these deposits. We can limit our analysis to only those bank-time observations where the deposit share from the counterparty in question represents at least 10%. Results are depicted in [Appendix Table A.3](#) and are highly similar.

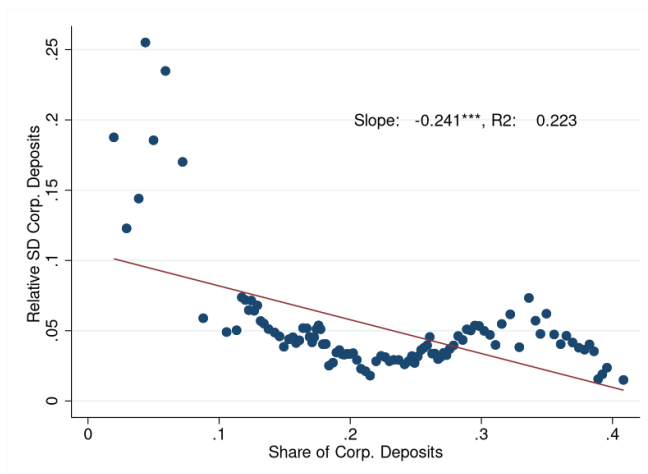
However, it is worth emphasizing that the negative relationships between the share of a bank's deposits that are comprised of flighty depositors and their actual flightiness does not overcome aggregate differences in volatility across depositor types. Banks that cater to more volatile depositors still face greater variability and greater monthly draw-downs on average. However, the ability for banks to better manage depositor types they specialize in serving may have implications for how growth in different deposits is allocated to productive firms. We discuss this in detail in the following section.

3.3 Interest Rates by Bank-Types

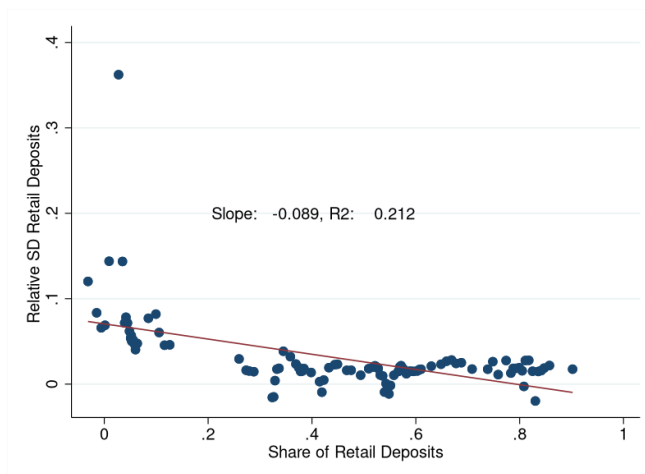
We do not have interest rate data for individual depositor-types. However, we can use aggregate Y-9C data to show coarse correlations between aggregate interest rates paid on deposits and the share of a bank's deposits derived from various counter-parties. In [Figure 5](#), we split banks in our sample by whether they receive the plurality of their deposits from Corporate-/ NBFI counterparties or Retail depositors. We make use of two groups for convenience. We can see that banks with more flighty deposits pay close to the federal funds rate. They aggressively reprice the amount they pay when the

Figure 4: Share of Deposits on Std. Dev. of Deposits – By Deposit Type

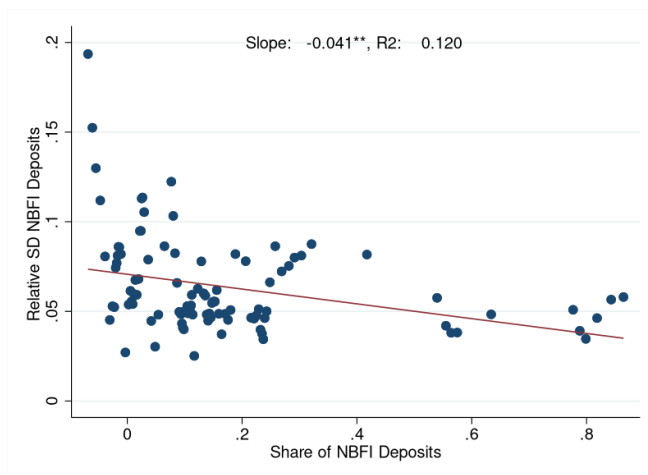
(a) Corp. Deposits



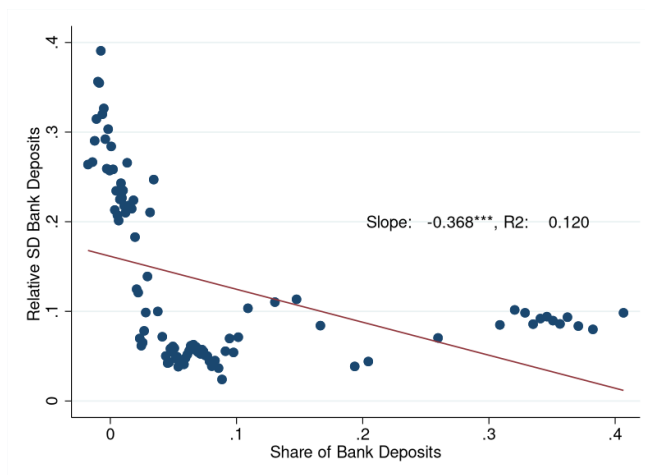
(b) Retail Deposits



(c) NBFI Deposits



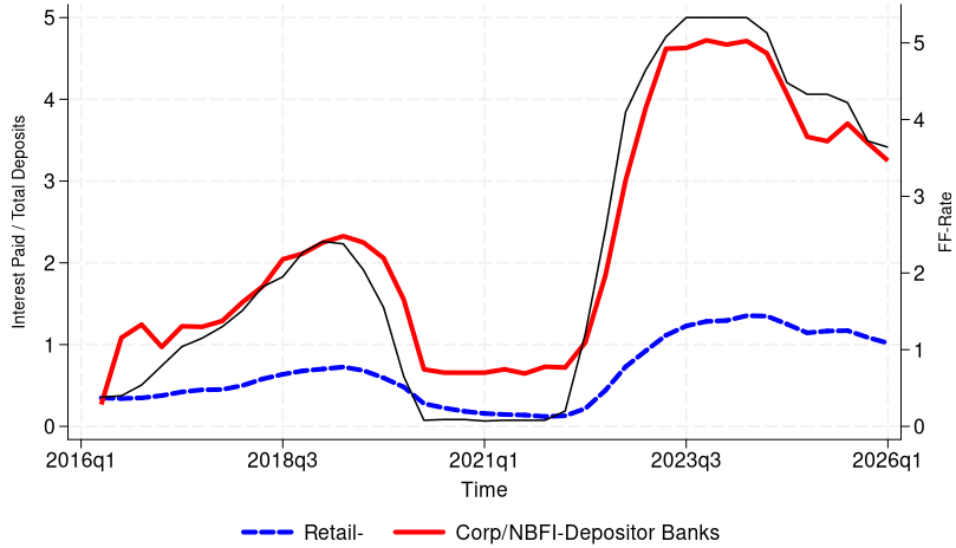
(d) Interbank Deposits



Notes: This figure shows binscatter plots relating the share of a bank's deposits originating from either NBFI, Corporate, or retail depositors to the relative standard deviation (12-month rolling standard deviation/ 12-month mean deposit base) of their respective deposit base. We make use of bank*month data and each bin contains at least 5 observations.

federal funds rate changes. Retail banks, on the other hand, pay a lot less for deposits on average and reprice more slowly following aggregate rate changes.

Figure 5: Interest Rate Paid – By Bank Type



Notes: We depict interest rates paid by bank (total interest expenses/total deposits) for banks in our sample. We split banks into two groups based on whether they receive the plurality of their deposits from corporate or NBFI counterparties or retail depositors (representing the same split as in Figure 1, above). We include the Federal funds rate on the secondary Y-axis.

We can corroborate these findings using a series of regressions that make use of the actual share of deposits derived from a given counterparty instead of binary cutoffs. In Table 2 we relate the rate paid for all deposits to the share of deposits that are either retail or corporate. We can see that even if we include bank fixed effects, higher shares of retail deposits are associated with lower rates paid on deposits.

In columns (5)-(8) we interact the share of deposits with the federal funds rate. We can see that the correlation between the federal funds rate and the aggregate interest paid on deposits is positive only for higher shares of corporate deposits and unrelated or negative for higher shares of retail deposits. While retail banks face other fixed costs related to their operations (such as a branch network) they pay significantly lower rates and showcase less sensitivity to aggregate rate changes.

3.4 Depositor Responses to Shocks

We can extend papers such as [Chen et al. \(2024\)](#) or [Canals-Cerdá \(2022\)](#) and use our proprietary data to analyze the responsiveness of different depositor types to drops in bank RoA. We compute a negative RoA "shock" as an event wherein a bank's RoA decreases (relative to the previous quarter) by more than two standard deviations (relative to the past 12 months). These large RoA shocks occur for around

Table 2: Interest Rate on Deposit Shares

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Interest Rate							
Share Retail Deposits	-1.109*** [0.149]		-1.234** [0.560]		-0.033 [0.209]		0.502 [0.546]	
Share Corp. Deposits		1.375*** [0.273]		0.788* [0.471]		-0.394 [0.380]		-0.788* [0.474]
Share Retail $\times$ Fed Funds					-0.595*** [0.083]		-0.496*** [0.043]	
Share Corp. $\times$ Fed Funds						1.078*** [0.165]		0.849*** [0.085]
Fixed Effects	Time	Time	Bank, Time	Bank, Time	Time	Time	Bank, Time	Bank, Time
Controls				Bank Size				
R ²	0.18	0.16	0.79	0.79	0.23	0.2	0.82	0.81
N	1,009	1,009	1,009	1,009	1,009	1,009	1,009	1,009

Notes: In this table we relate the aggregate interest rate paid by a bank for all deposits to the share of deposits that are obtained from either corporate or retail depositors. We include fixed effects as indicated. In columns (5)-(8) we interact our variable of interest with the federal funds rate. Standard errors are robust and clustered at the industry*time level while *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

5% of our bank-quarter observations. We can relate these shocks to logged depositor changes – after accounting for bank assets, capital, risk weighted assets, etc. Results are reported in Appendix Table A.4. Retail depositors are not very sensitive to these shocks⁴. Corporate depositors and especially NBFI depositors are responsive to RoA shocks. these customers – mostly uninsured – seem the most susceptible to poor bank returns. We extend the analysis somewhat by interacting our RoA-shock variable with indicators for whether a bank is oriented towards, NBFI-, corporate-, or retail depositors. We find that depositors of such specialized banks are somewhat less – not more – responsive to these shocks. However, these differences are small and not statistically significant in the face of our controls.

Finally, an argument can be made that different bank types, which cater to different types of depositors, also use deposits to differing extents. If NBFI banks do not fund themselves with deposits, then an analysis of how their lending corresponds to their deposit structure is pointless. However, as can be seen from Appendix Figure A.6, each of the bank types is surprisingly similar on average. Differences within bank types are far greater than across. While Foreign banks tend to be NBFI-oriented, they are not the only NBFI-depositor oriented banks.

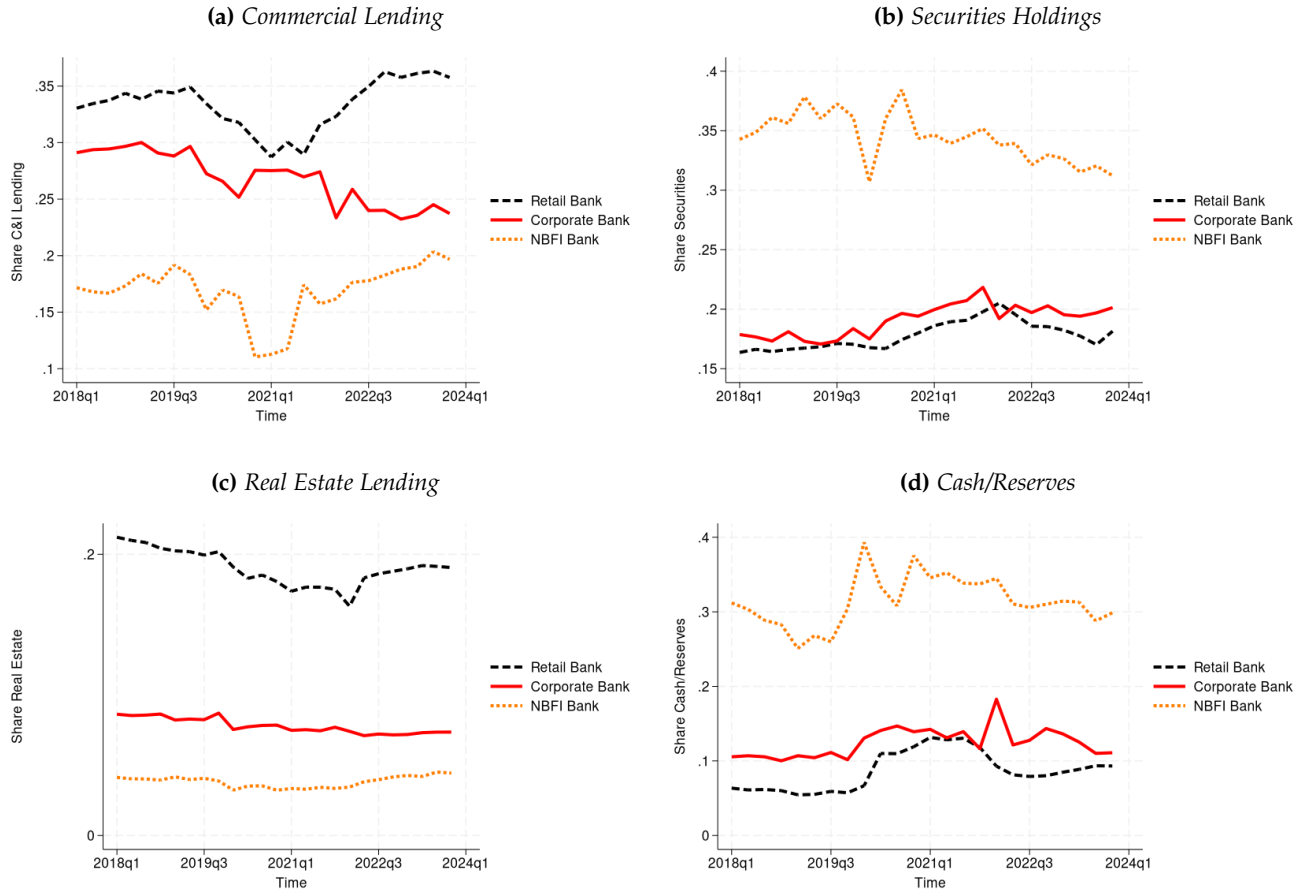
⁴This corroborates earlier evidence by [Blickle et al. \(2024\)](#) who show that retail depositors fail to run until news of a bank's insolvency becomes public.

4 Matching Assets to Deposit Structure

4.1 Aggregate Bank-Level Results

Different bank-types engage in different degrees of maturity transformation. This is easily depicted in Figure 6, where we use the same bank-type definitions, based on which depositor-types constitute the largest depositor group, as in Figure 1 above. We plot the share of a bank's balance sheet directed towards C&I lending, real estate lending, securities, or cash and reserves held. One can see that retail-depositor banks actually engage in more C&I lending, on average, than corporate- or NBFI-oriented banks do. Moreover, they devote a far greater share of their assets to real estate lending that remains on their balance sheet (primarily jumbo mortgages or commercial real estate). NBFI banks, on the other hand, are almost entirely concentrated into cash/reserves or other securities. If we break down securities further, we find that NBFI banks focus almost exclusively on treasuries (not reported for brevity).

Figure 6: *Bank Types and Lending Behavior*



Notes: this Figure depicts the share of a bank's balance sheet directed towards different types of investments. Panel (a) shows C&I lending, including undrawn credit lines, Panel (b) shows securities held by banks, Panel (c) shows real estate lending (commercial and residential) and panel (d) shows Cash/Reserves held.

We can analyze the relationships between a bank's balance sheet composition and its assets, discussed above, more directly. In Table 3 we show separate regressions that relate a bank's C&I lending, securities held, real estate, or cash/reserves to the share of its deposits that are from retail, corporate, or NBFI depositors, the share that are demandable (vs. time deposits of any maturity), or the share that are insured. One can see that retail and insured deposits are related to a higher share of real estate lending. On average, short-term deposit banks and corporate deposit banks have a higher share of C&I lending, though short term depositor banks and corporate depositor oriented banks tend to match their deposits with revolving credit lines of less than 1 year maturity (not reported for brevity). Banks with high retail and banks with high levels of insured deposits are less likely to invest in liquid, short term securities or cash.

If we include bank fixed effects (columns (5) -(8)), we are ostensibly analyzing a responsiveness to a change in a certain deposit-share, relative to a bank's average. Higher retail and insured deposit shares are again correlated with more real estate lending and lower cash holdings. However, we do find that these deposit shares are now also correlated with C&I lending. Conversely, C&I lending is no longer positively associated with higher relative shares of corporate deposits. This is some initial evidence that not all banks treat different deposits in the same way. Unsolicited (and likely volatile) *changes in* corporate deposits may not be invested in C&I or real estate lending. In the Appendix we run the same analysis on a slightly larger sample of banks where we can match Y9C and deposit data but not Y14 data. In these regressions we additionally include controls for a bank's tier 1 equity relative to total assets and its risk weighted assets relative to total assets. As can be seen in Appendix table A.6, the results are indistinguishable.

Another dimension to observe, when looking at the impact of deposits on lending, would be to look at changes in deposits directly. In Table 4, we look at the effect of changes in corporate, retail or NBFI deposits on changes in C&I, real estate, cash or securities lending in the subsequent quarter. In columns (1)-(4) results match expectations. Increases in corporate deposits are invested in all types of lending. Retail deposit increases are invested broadly too, though less into C&I lending than one might expect, given aggregate levels of C&I lending for retail-oriented banks. Finally, NBFI deposits – which are the most volatile on average – are invested primarily into more liquid cash/reserves or securities.

In columns (5)-(8) we include not only bank fixed effects, to control for average levels, but we also interact our variable of interest with the share of deposits a bank has issued to certain types of depositors. We now see that corporate deposit growth is invested into C&I lending primarily by those banks that

Table 3: Balance Sheet Composition on Deposit Shares

	(1) C&I Lending Share	(2) Securities Share	(3) Cash/Reserves Share	(4) Real Estate Share	(5) C&I Lending Share	(6) Securities Share	(7) Cash/Reserves Share	(8) Real Estate Share
Panel A								
Share of Retail Deposits	0.056* [0.029]	-0.221*** [0.012]	-0.289*** [0.012]	0.064*** [0.020]	0.407*** [0.059]	-0.131*** [0.039]	-0.433*** [0.054]	0.170*** [0.031]
Panel B								
Share of Corp. Deposits	0.884*** [0.057]	-0.179*** [0.035]	-0.284*** [0.039]	0.521 [0.342]	-0.294*** [0.067]	0.195*** [0.043]	0.593*** [0.057]	-0.332*** [0.031]
Panel C								
Share of NBFI Deposits	-0.220*** [0.028]	0.245*** [0.011]	0.342*** [0.009]	-0.231*** [0.018]	-0.036 [0.092]	0.081 [0.059]	0.055 [0.086]	-0.037 [0.047]
Panel D								
Share of ST-Deposits	0.301*** [0.065]	0.395*** [0.030]	0.330*** [0.037]	0.050 [0.046]	-0.204*** [0.035]	0.093*** [0.023]	0.157*** [0.033]	-0.002 [0.019]
Panel E								
Share of Insured Deposits	0.012 [0.033]	-0.205*** [0.015]	-0.327*** [0.014]	0.134*** [0.022]	0.217*** [0.052]	-0.035 [0.034]	-0.360*** [0.046]	0.157*** [0.026]
N	500	500	500	500	500	500	500	500
Fixed Effects	-	-	-	-			Bank	

Notes: This Table shows the relationship between a bank's asset-side composition (the share of a balance sheet invested in C&I lending (including commitments), value of securities held relative to all assets, the cash or reserves held relative to all assets and real estate lending (including commercial and residential) relative to all assets) and depositor composition (the share of depositors that are insured, the share that are short termist or demandable, the share that are corporations and the share that are retail). Each cell represents an individual and independent regression. Standard errors are heteroskedasticity robust while *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

already have a large share of corporate deposits. Similarly, retail-banks are more likely to invest in real estate. Perhaps a little surprisingly, these banks are also considerably more likely to hold retail deposits in cash. This may be a function of the fact that it takes time for real estate opportunities to materialize, necessitating a temporary cash storage. NBFI banks focus newly arriving NBFI deposits into cash while all other banks concentrate NBFI deposits into securities. Neither invest in more long-term loans. The results for short-term deposits look similar to corporate depositors and the results for insured depositors mirror retail depositors (not shown for brevity).

4.2 Loan-Level Results

4.2.1 Funding Structure and Loan Terms

Table 4: Lending Changes on Deposit Changes

	(1) Δ C&I Lending	(2) Δ Securities	(3) Δ Cash/Reserves	(4) Δ Real Estate	(5) Δ C&I Lending	(6) Δ Securities	(7) Δ Cash/Reserves	(8) Δ Real Estate
Panel A								
Δ Corp. Dep.	0.122*** [0.031]	0.039** [0.018]	1.006*** [0.120]	0.172*** [0.041]	-0.012 [0.046]	0.023 [0.025]	0.200 [0.228]	0.239*** [0.079]
Share Corp. Dep.					-0.140*** [0.047]	-0.031 [0.026]	-0.048 [0.126]	-0.136*** [0.044]
Δ Corp. Dep. * Share Corp. Dep.					0.745*** [0.178]	0.124 [0.124]	4.553*** [1.107]	-0.346* [0.152]
Panel B								
Δ Retail Dep.	0.001 [0.038]	0.109*** [0.024]	0.184* [0.094]	0.151*** [0.029]	0.052* [0.023]	-0.002 [0.030]	-0.004 [0.105]	0.042 [0.031]
Share of Retail Deposits					0.027 [0.024]	0.013 [0.015]	-0.039 [0.052]	-0.083*** [0.015]
Δ Retail Dep. * Share of Retail Deposits					-0.340 [0.292]	0.709*** [0.117]	2.057*** [0.520]	1.116*** [0.154]
Panel C								
Δ NBFI Dep.	0.023 [0.038]	0.117*** [0.021]	0.140* [0.074]	0.051 [0.044]	0.005 [0.046]	0.110*** [0.024]	-0.014 [0.079]	0.065** [0.026]
Share of NBFI Deposits					0.001 [0.031]	-0.017 [0.016]	-0.063 [0.049]	0.080*** [0.016]
Δ NBFI Dep. * Share of NBFI Deposits					0.119 [0.163]	0.084 [0.148]	2.299*** [0.484]	-0.189 [0.158]
N	475	475	475	475	475	475	475	475
Fixed Effects	Time				Time, Bank			

Notes: This table shows quarterly log changes in a bank's investments (the amount invested in C&I lending (including commitments), the value of securities held, the cash or reserves held, and real estate lending (including commercial and residential)) relative to contemporaneous changes in corporate, retail, or NBFI deposits. Columns (5) -(8) include bank fixed effects and interactions with the share of a bank's balance sheet the corresponding depositor-type represents. Standard errors are heteroskedasticity robust while *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Y-14 Schedule H allows us to examine how the terms of individual C&I loans (i.e. rates, maturities, and sizes) vary with the funding structures of the issuing banks. Given the high number of observations, we are able to account for loan-specific characteristics, a bank's assessment of borrower risk, collateral type, and other fixed effects, which ensures that we are comparing like loans with like loans.

We first link loan terms directly to the share of deposits from corporate, retail, and NBFI sources in Table 5. This follows the analysis pattern we established above. The results show stark differences across banks with varying depositor type preferences. Banks with a larger corporate (or NBFI) deposit share charge significantly higher loan rates, extend substantially shorter maturities, and make smaller loans. In contrast, banks with greater retail or insured deposit shares (not reported for brevity) offer lower rates, longer maturities, and larger loans. The economic magnitudes are considerable: moving from the 25th to the 75th percentile in retail deposit share is associated with a maturity increase of roughly seven quarters. These patterns are consistent with the view that stable funding enables banks to extend more generous credit terms, while volatile funding leads to more restrictive lending.

Table 5: Loan Terms on Deposit Shares

	(1) Rate	(2) Maturity	(3) Log Size	(4) Rate	(5) Maturity	(6) Log Size	(7) Rate	(8) Maturity	(9) Log Size
Share Corp. Dep.	1.266*** [0.026]	-11.668*** [0.110]	-1.199*** [0.012]						
Share Retail Dep.				-0.313*** [0.014]	6.969*** [0.059]	1.217*** [0.007]			
Share NBFI Dep.							0.145*** [0.0013]	-4.840*** [0.054]	-1.019*** [0.006]
Fixed Effects	Collateral type, industry*year*quarter, loan purpose, loan type								
Controls	Total interest paid, bank assets, bank expenses, loan terms (other than the dep. var), loan rating								
R ²	0.26	0.32	0.28	0.26	0.32	0.28	0.26	0.32	0.28
N	3,339,792	3,339,792	3,339,792	3,339,792	3,339,792	3,339,792	3,339,792	3,339,792	3,339,792

Notes: This Table relates loan terms (interest rate, maturity remaining, and log loan size) to the degree to which a bank makes use of a certain type of deposit. We include bank-controls such as interest paid on all loans, banks operating expenses, and total assets as well as loan controls such as the rate, the size, and the maturity (in those regressions in which they are not the dependent variable). We further include fixed effects such as the time*industry, the collateral type, the loan type, the loan purpose, the bank's internal rating of the borrower's risk (standardized across banks). Standard errors are robust and clustered at the industry*time level while *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

These results corroborate earlier work – either theoretical or based on public data – which has argued that retail banks with lower deposit betas charge less for some loans while corporate depositor oriented banks have to pay higher rates to compete with money-market-funds. See for instance: [Craig and Koepke \(2019\)](#), [Dräger et al. \(2021\)](#), [Ritz and Walther \(2020\)](#), [Drechsler et al. \(2017\)](#), [Hannan and Berger \(1991\)](#), [Chodorow-Reich et al. \(2022\)](#), [d'Avernas et al. \(2023\)](#), or [Drechsler et al. \(2021\)](#). Or [Carletti et al. \(2024\)](#) who show that households are less likely to move deposits even in a run.

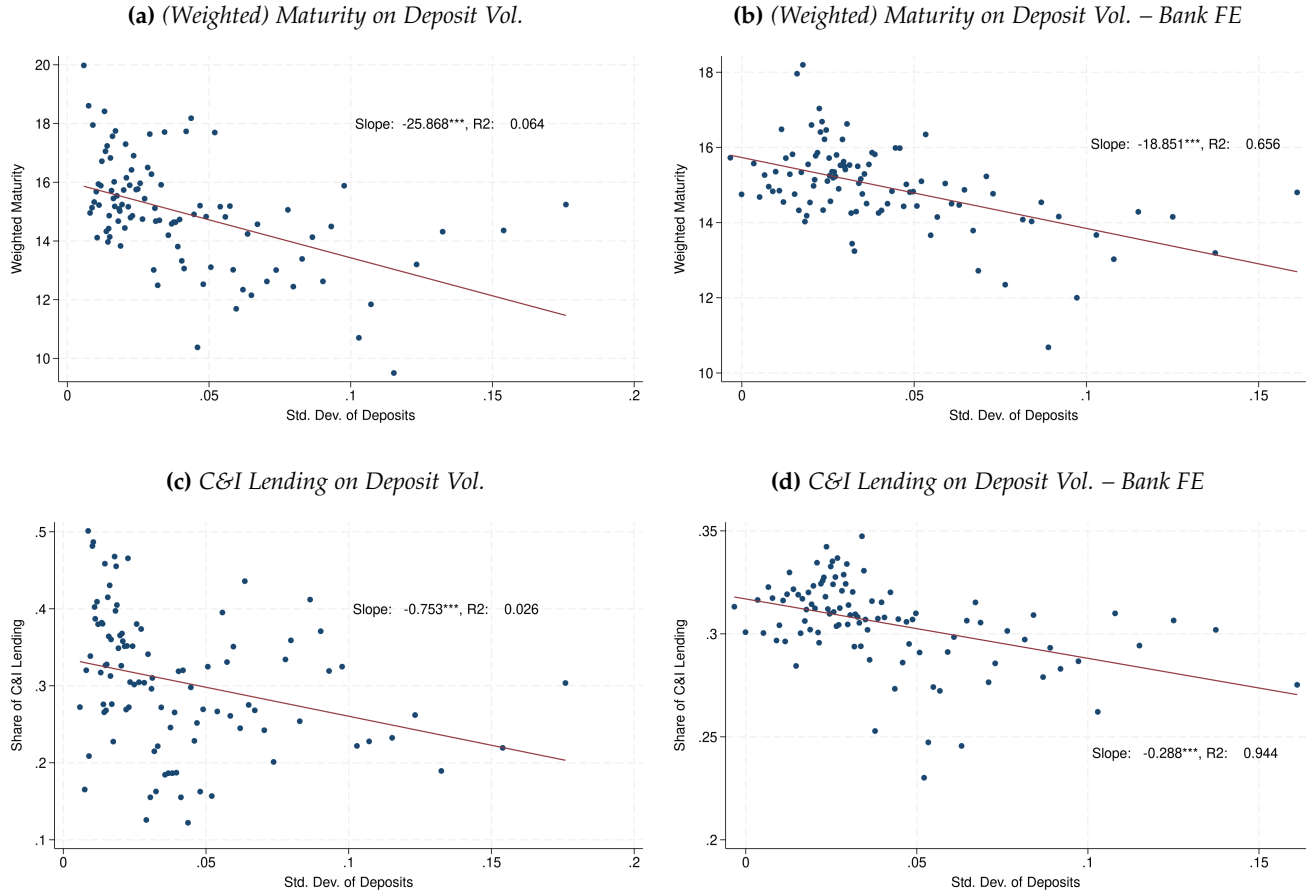
These results are perhaps initially somewhat surprising, given the fact that corporate-deposit banks have a relatively high C&I lending share on aggregate (see Table 3). However, the results discussed in the section above do not broach the topic of terms. Instead, we explicitly see lower maturity loans for more corporate deposit heavy banks. While banks with more volatile deposit bases charge higher rates for smaller loans, it does not necessarily imply that they make fewer such loans in aggregate. Overall, if other banks are focused on different lending activities, corporate-depositor-oriented banks may still be the only viable option for some C&I borrowers.

4.2.2 Depositor Volatility and Lending

In Figure 7, one can see that the average maturity of C&I loans made by banks with more volatile deposits is lower. We relate the relative standard deviation of deposits (measured as the 12 month

standard deviation relative to the 12 month mean level of assets) to the average loan maturity. We weight loans by their (drawn) size when computing average maturity. The observed negative relationship holds across and within banks (Panel (b)), meaning that banks experiencing an uptick in deposit volatility opt for lower maturity loans. Moreover, these results hold despite the fact that banks with a more volatile depositor base engage in less C&I lending to begin with (Panels (c) and (d)).

Figure 7: C&I Lending and Loan Maturity vs. Deposit Volatility



Notes: This Figure shows the relationship between average C&I loan maturity, weighted for loan size, in quarters and the relative standard deviation of a bank's deposits (12-month rolling standard deviation over 12 month mean deposit level) in Panels (a) and (b). Panels (c) and (d) depict the relationship between C&I lending (including credit line commitments) and the relative standard deviation of deposits. Panels (b) and (d) include bank fixed effects. Each bin contains at least 5 bank-time observations.

In Table 6 we show that the average variability of a bank's total deposits is related to the loan terms it offers – much as we might have expected, given the aggregate results discussed above. More volatile deposits are associated with charging higher rates for shorter maturity loans that may also be smaller. This holds even if we include the ratings a bank assigns to loans. The effect is pronounced; a one standard deviation increase in the relative variability of deposits is associated with a 6 basis points increase in the interest rate charged, all else equal.

Table 6: Loan Terms on Deposit Variability

	(1) Rate	(2) Maturity	(3) Log Size	(4) Rate	(5) Maturity	(6) Log Size
Relative Std. Dev. total deposits	0.194*** [0.062]	-4.835*** [0.260]	-0.747*** [0.029]	0.357*** [0.063]	-4.912*** [0.276]	-0.732*** [0.033]
Fixed Effects I	Collateral type, industry*year*quarter, loan purpose, loan type					
Fixed Effects II	Loan Rating					
Controls	int. paid, bank assets, bank expenses, loan terms					
R ²	0.26	0.32	0.28	0.26	0.33	0.28
N	3,220,902	3,220,902	3,220,902	3,220,902	3,220,902	3,220,902

Notes: This Table relates loan terms (interest rate, maturity remaining, and log loan size) to the variability of a bank's total deposits (measured as σ/μ .) We interact our variable of interest with bank-type dummies. We include bank-controls such as interest paid on all loans, banks operating expenses, and total assets as well as loan controls such as the rate, the size, and the maturity (in those regressions in which they are not the dependent variable). We further include fixed effects such as the time*industry, the collateral type, the loan type, the loan purpose, the bank's internal rating of the borrower's risk (standardized across banks). Standard errors are robust and clustered at the industry*time level while *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Table 7 shows that the relationship between funding volatility and lending behavior differs across bank types. Given the similarities between NBFI and corporate-depositor-oriented banks, as well as the similarities between insured-depositors and retail-depositor-oriented banks, we focus only on two types of entities—corporate banks and retail-depositor-oriented banks—in these tables for convenience.

Retail banks respond to greater depositor volatility with higher rates and shorter maturities and smaller loans, suggesting a precautionary tightening when their traditionally stable base becomes less predictable. Corporate banks, by contrast, increase maturities and loan sizes in response to volatility. This may simply be a reflection of the fact that such banks are more familiar with how to value and invest in less stable deposits. Compared to less corporate-depositor-oriented banks, this familiarity with volatility expresses itself as *relatively* more generous lending.⁵

4.2.3 Deposit Changes and Loan Terms

Taken together, the loan-level results discussed thus far reveal that depositor composition affects lending terms. Moreover, the sensitivity of those terms to funding volatility is pronounced. Stable funding sources, such as retail and insured deposits, are generally associated with more favorable credit terms, while volatile sources, such as corporate deposits, correspond to tighter lending. However, different banks – with different depositor specializations – react differently to deposit variability.

⁵Given that deposit volatility rose in part due to an increase in deposit volume – as discussed below – , it would naturally follow that experienced, corporate-oriented banks may use these funds to engage in C&I lending.

Table 7: Loan Terms on Deposit Variability Interacted with Bank Type

	(1) Rate	(2) Maturity	(3) Log Size	(4) Rate	(5) Maturity	(6) Log Size
Interaction: Std. Dev. *Retail share	19.148*** [0.422]	-15.074*** [1.760]	-7.215*** [0.197]			
Relative Std. Dev. total deposits	-9.461*** [0.220]	-10.433*** [0.917]	3.201*** [0.103]	5.733*** [0.218]	-42.324*** [0.909]	-4.154*** [0.102]
Retail share	-1.142*** [0.023]	6.273*** [0.097]	1.523*** [0.011]			
Interaction: Std. Dev. *Corporate share				-20.776*** [0.807]	139.103*** [3.364]	12.571*** [0.379]
Corporate share				2.035*** [0.040]	-16.900*** [0.165]	-1.679*** [0.019]
Fixed Effects	Collateral type, industry*year*quarter, loan purpose, loan type					
Controls	int. paid, bank assets, bank expenses, loan terms, loan rating					
R ²	0.26	0.32	0.28	0.26	0.33	0.28
N	3,220,902	3,220,902	3,220,902	3,220,902	3,220,902	3,220,902

Notes: This Table relates loan terms (interest rate, maturity remaining, and log loan size) to the variability of a bank's total deposits (measured as σ/μ .) We interact our variable of interest with bank-type dummies. We include bank-controls such as interest paid on all loans, banks operating expenses, and total assets as well as loan controls such as the rate, the size, and the maturity (in those regressions in which they are not the dependent variable). We further include fixed effects such as the time*industry, the collateral type, the loan type, the loan purpose, the bank's internal rating of the borrower's risk (standardized across banks). Standard errors are robust and clustered at the industry*time level while *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

A final concept worth considering is the relationship between deposit changes and loan terms. Table 8 shows that increases in corporate deposits are associated with higher loan rates, shorter maturities, and modestly smaller loans, suggesting that such inflows are viewed as temporary or higher-cost funding. The effects are sizeable; a one standard deviation increase in "expensive" corporate deposits at the average bank in our sample would result in a 2 basis points increase in the cost of a loan (relative to an unaffected bank). Retail deposit inflows, by contrast, are linked to lower rates, longer maturities, and slightly larger loans, reflecting the perception of these funds as stable and well-suited for term lending.

Tables 9 and 10 again show that the effects of deposit growth are heterogeneous across banks with different familiarities with certain depositor types. In columns (1)-(3) we interact the share of a bank's depositor base that stems from retail depositors with the change in corporate or retail deposits respectively. In columns (4)-(6) we interact our changes in deposits with the degree to which a bank relies on corporate deposits. We see that retail-depositor oriented banks are more likely to raise rates and reduce loan size/maturities in response to rising corporate deposits. Conversely, corporate-depositor-

oriented banks are more comfortable lowering rates and increasing loan sizes/maturities in response to an increase in corporate deposits. Familiarity with the deposit type makes it more likely that the affected bank will lend newly arriving deposits to customers.

We make use of retail deposit inflows in Table 10. We find that retail-oriented banks again show little inclination to lend newly arriving retail deposit funds to C&I borrowers (preferring residential real estate investments instead). Interestingly, corporate deposit oriented banks are similarly disinclined to invest retail deposits in C&I lending, preferring even longer term investments such as real estate.

Table 8: *Loan Terms on Deposit Changes*

	(1) Rate	(2) Maturity	(3) Log Size	(4) Rate	(5) Maturity	(6) Log Size
Δ Corp. Deposits	0.458*** [0.029]	-0.713*** [0.120]	-0.063*** [0.014]			
Δ Retail. Deposits				-0.454*** [0.049]	2.286*** [0.201]	0.063*** [0.023]
Fixed Effects	Collateral type, industry*year*quarter, loan purpose, loan type					
Controls	int. paid, bank assets, bank expenses, loan terms, loan rating					
R ²	0.26	0.32	0.28	0.26	0.34	0.3
N	3,314,961	3,314,961	3,314,961	3,339,792	3,339,792	3,339,792

Notes: This Table relates loan terms (interest rate, maturity remaining, and log loan size) to the log changes in certain depositor types. We interact our variable of interest with bank-type dummies. We include bank-controls such as interest paid on all loans, banks operating expenses, and total assets as well as loan controls such as the rate, the size, and the maturity (in those regressions in which they are not the dependent variable). We further include fixed effects such as the time*industry, the collateral type, the loan type, the loan purpose, the bank's internal rating of the borrower's risk (standardized across banks). Standard errors are robust and clustered at the industry*time level while *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Of course, it is worth noting that we lack a way of identifying exogenous deposit growth in the above analyses. It is entirely possible that corporate banks attracted corporate depositors because these banks had lending opportunities. Similarly, it is possible that deposit growth and lending growth occurred together as part of a package deal. In the next section, we make use of the COVID shock as a source of exogenous deposit variation.

Table 9: Loan Terms on Deposit Changes Interacted with Depositor Shares (Corporate)

	(1) Rate	(2) Maturity	(3) Log Size	(4) Rate	(5) Maturity	(6) Log Size
Interaction: Δ Corp. Dep * Retail Share	5.512*** [0.181]	4.488*** [0.747]	-0.483*** [0.084]			
Δ Corp. Dep	-2.427*** [0.092]	-1.992*** [0.382]	0.262*** [0.043]	1.638*** [0.094]	1.994*** [0.391]	-0.159*** [0.044]
Retail Share	-0.153** [0.068]	-6.713*** [0.283]	0.407*** [0.032]			
Interaction: Δ Corp. Dep * Corp. Share				-4.744*** [0.344]	-8.603*** [1.432]	0.942*** [0.162]
Corp. Share				1.411*** [0.028]	-11.458*** [0.117]	-0.809*** [0.037]
Fixed Effects	Collateral type, industry*year*quarter, loan purpose, loan type					
Controls	int. paid, bank assets, bank expenses, loan terms, loan rating					
R ²	0.26	0.34	0.3	0.26	0.33	0.3
N	3,314,961	3,314,961	3,314,961	3,314,961	3,314,961	3,314,961

Notes: This Table relates loan terms (interest rate, maturity remaining, and log loan size) to the log changes in corporate deposits. We interact our variable of interest with the share of a bank's deposits originating from either retail or corporate depositors. We include bank-controls such as interest paid on all loans, banks operating expenses, and total assets as well as loan controls such as the rate, the size, and the maturity (in those regressions in which they are not the dependent variable). We further include fixed effects such as the time*industry, the collateral type, the loan type, the loan purpose, the bank's internal rating of the borrower's risk (standardized across banks). Standard errors are robust and clustered at the industry*time level while *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

4.3 Extension: Loan Risk

Before addressing the endogeneity issue head on, it is first worth addressing two related issues as extensions to our baseline results. Specifically, it is worth analyzing loan and bank risk, something we have thus far controlled but not analyzed explicitly. We first show – in Appendix Table A.5 – that banks with higher shares of more stable deposit lend to more risky borrowers. We can ascertain this by looking at the rating assigned to loans, where the rating system runs from 1-10 with lower numbers being better ratings. Banks with high shares of retail deposits have slightly though significantly worse rated loans, all else equal. In a similar vein, we see that loans by retail-depositor oriented banks perform worse (i.e. become non-accruing) more often. In short: these banks and their placid depositors are more willing and able to take risk.

Commensurate with this assessment, we can see from Appendix Figure A.7 that the risk weighted assets of banks with more stable funding sources are higher. Higher RWA naturally come with having to hold more/larger (expensive) capital buffers. Moreover, we can see from Appendix Figure A.8 that

Table 10: Loan Terms on Deposit Changes Interacted with Depositor Shares (Retail)

	(1) Rate	(2) Maturity	(3) Log Size	(4) Rate	(5) Maturity	(6) Log Size
Interaction:Δ Ret. Dep * Retail Share	5.414*** [0.282]	-9.463*** [1.174]	-0.293** [0.135]			
Δ Retail Dep	-2.619*** [0.136]	6.365*** [0.568]	0.034 [0.065]	-0.939*** [0.153]	23.243*** [0.639]	0.438*** [0.073]
Retail Share	-0.392*** [0.015]	7.063*** [0.062]	0.395*** [0.032]			
Interaction:Δ Ret. Dep * Corp. Share				3.089*** [0.563]	-78.554*** [2.346]	-1.358*** [0.266]
Corp. Share				1.207*** [0.028]	-10.169*** [0.118]	-0.772*** [0.036]
Fixed Effects	Collateral type, industry*year*quarter, loan purpose, loan type					
Controls	int. paid, bank assets, bank expenses, loan terms, loan rating					
R ²	0.26	0.32	0.3	0.26	0.32	0.3
N	3,339,792	3,339,792	3,339,792	3,339,792	3,339,792	3,339,792

Notes: This Table relates loan terms (interest rate, maturity remaining, and log loan size) to the log changes in retail deposits. We interact our variable of interest with the share of a bank's deposits originating from either retail or corporate depositors. We include bank-controls such as interest paid on all loans, banks operating expenses, and total assets as well as loan controls such as the rate, the size, and the maturity (in those regressions in which they are not the dependent variable). We further include fixed effects such as the time*industry, the collateral type, the loan type, the loan purpose, the bank's internal rating of the borrower's risk (standardized across banks). Standard errors are robust and clustered at the industry*time level while *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

banks with more retail depositors have slightly higher general expenses – perhaps due to maintaining a larger branch network.

Finally, while retail depositors may be cheaper than NBFID deposits, the costs of maintaining a branch network, combined with higher costs related to loan risk mean that the profitability of different bank types is highly comparable. From Figure A.7, we can see that profitability is only marginally related to the share of deposits that are considered more flighty. While the relationship is slightly negative, it is generally insignificant.

4.4 Extension II: Risk Index – Asset Duration vs. Liability Maturity

As has become evident in the discussion thus far, banks generally align asset duration with liability maturity. Given that maturity transforming flighty deposits may be risky, this is managerially prudent. However, our results thus far beg the question of whether banks ever fail to align asset duration and

depositor maturity – and whether this presents as material risk.

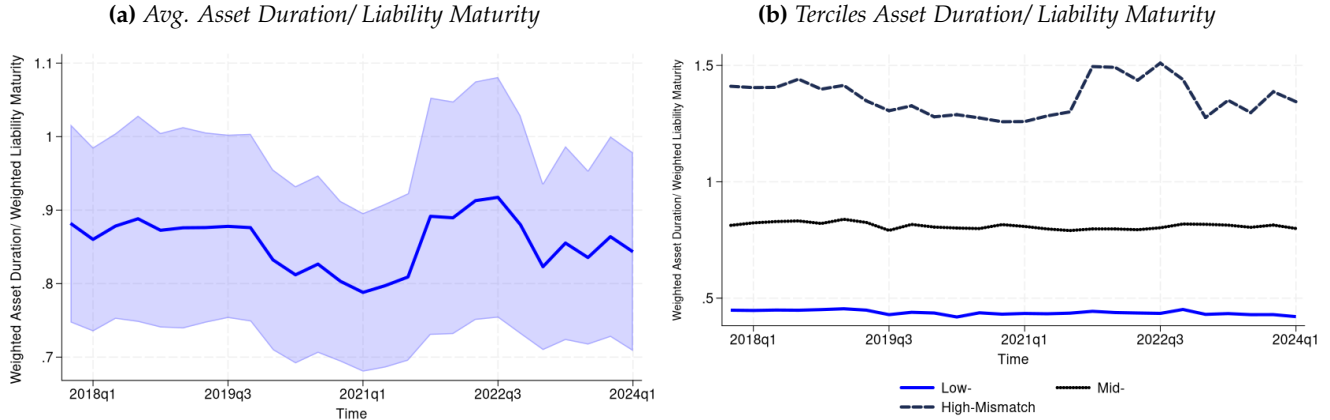
We can easily compute the weighted average maturity of deposits using our actual data. We calculate run-off rates for different types of depositors (retail insured, retail uninsured, brokered accounts, wholesale corporate, wholesale NBFI, etc.). We depict some of these results in the Appendix. We simplify our measures/calculations to make them comparable with LCR assumptions. During our sample period, many deposits were more stable than LCR assumptions might have suggested. This holds particularly when rates are low. However, as soon as rates began to rise, deposits, especially corporate and NBFI deposits became more flighty (see also discussions in [Blickle et al. \(2025\)](#)).

We can calculate asset duration with some assumptions. We make use of actual weighted loan maturity from Y14 data for a bank's C&I lending. Real estate portfolio duration is estimated by assigning different effective durations by loan type. Residential mortgages amortize and prepay, we therefore assume an effective duration of 5 years for residential real estate loans, consistent with standard estimates for agency MBS, and 7 years for commercial real estate loans, reflecting their longer balloon structures and weaker prepayment incentives. For the securities portfolio, we compute weighted duration from the Y-14 maturity and repricing distribution. Cash, fed funds sold, and other short-term assets are assigned a duration of 0.1 years, and consumer loans 2 years. The bank-level weighted asset duration is then the asset-share-weighted average across these categories.

Two caveats apply. First, our duration assignments are static and do not capture convexity or the endogenous lengthening of mortgage duration as rates rise — if anything, this biases us against finding large mismatches in 2022. Second, we deliberately ignore interest rate derivatives, both because hedging disclosures are coarse and because recent evidence suggests banks' swap positions offset only a small fraction of securities duration risk. We do however apply a 20% duration penalty to securities held to maturity vs those held as available for sale. This reflects real economic consequences related to repricing the former relative to the latter.

In Figure 8, we plot the asset duration scaled by the liability maturity for banks in our sample. We include the mean and the 90% confidence interval as a shaded blue area. Values larger than one represent a greater degree of risk as there is no "buffer" to facilitate deposit outflow. Overall mismatch fell during COVID and rose slightly with the start of rate hikes. The average has consistently been well below 1. Panel (b) of the Figure includes averages for the top, middle and bottom tercile of banks in our data base on January 2020 ratios. We can see that the majority of changes and the majority of the risk is concentrated in banks in the top tercile.

Figure 8: Asset Duration/ Liability Maturity



Notes: Panel (a) depicts asset duration relative to liability maturities, averaged for the banks in our sample. We include the 95% confidence interval as a shaded area. Asset durations are calculated using assumptions discussed in the text or outright maturity data where available (i.e. for Y14 C&I loans). Liability maturities are calculated using actual runoff data. Panel (b) breaks banks in our sample into three groups based on their asset duration/liability maturity.

In Figure 9 we plot asset durations and liability maturity for banks in our sample in a scatter plot. We make use of data for Q4 of 2022. We scale individual bubbles in the scatter based on bank 3 size groupings. We see that the majority of banks are well aligned with the idea that only banks with longer maturity liabilities should maturity transform. If anything, large banks in particular are conservative. The only outlier is SVB, which failed the following quarter in large part due to having failed to align depositor flightiness and asset duration. This highlight the fact that our measure is a useful representation of risk.

5 COVID-19 as a Source of Exogenous Deposit Variation

Deposits grew sharply during the COVID-19 pandemic as consumers saved money – due in part to a lack of spending opportunities –, and the government engaged in quantitative easing measures and outright relief spending. Some of the deposit growth can be inferred from Figure 1, Panel (d). We can look at the growth in deposits held by various counter-parties more directly in Figure 10. In Panel (a), we index each type of deposit (Corporate, NBFI, and Retail) to January of 2019. Here, we can see that the deposits of corporations grew most strongly during the entire period, before beginning a longer contraction at the very end of the COVID/lockdown cycle. NBFI deposits spiked and contracted initially (as QE was implemented), before growing more slowly during the remaining period. Finally, retail deposits grew almost steadily throughout the period. In Panel (b) we show the distribution of logged

Figure 9: Asset Duration/ Liability Maturity – Scatter Q4 2022



Notes: This Figure depicts a scatterplot of Asset Durations and Liability Maturities for banks in our data. Asset durations are calculated using assumptions discussed in the text or outright maturity data where available (i.e. for Y14 C&I loans). Liability maturities are calculated using actual runoff data. Bank sizes are indicated (in bins) by bubble sizes. SVB is highlighted as it constituted a clear outlier before failure.

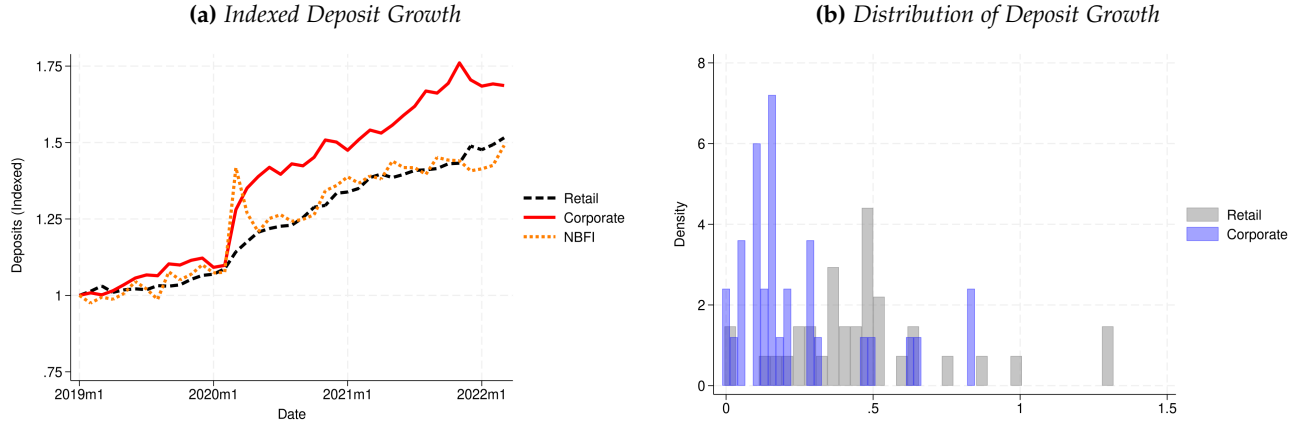
deposit growth for banks in our sample over the entire COVID period. One can see that there exists significant heterogeneity in the degree to which different banks grew their deposit base during COVID.⁶

Importantly, for our purposes, the deposit growth was in part related to existing customer relations. Corporations and retail consumers deposited funds with existing bank partners. A very significant portion of this deposit growth was exogenous to bank lending opportunities during the crisis. We can view the deposit growth during the COVID-period as quasi exogenous; it provides a useful natural experiment to study the lending decisions of different types of banks.

We begin by examining how deposit changes during COVID affected aggregate bank lending behavior in Table 11. While inflows of corporate deposits were not associated with a systematic increase in C&I lending at the aggregate level, we observe heterogeneity across banks (just as we do above). Specifically, banks with a pre-existing specialization in corporate depositors were somewhat more likely to increase C&I lending in response to corporate deposit growth. This effect is economically large. We may lack sufficient observations given the large variability of our variables in question, so the effects are borderline significant. Corporate banks also direct some of these (usually more volatile) corporate deposits towards real estate lending. Possibly because loan demand was high and corporate banks felt comfortable in being able to manage their deposit base enough to perform long-term maturity

⁶Different banks were observed to grow at different rates in 2023 as well. There was a flight to large banks during that time. See for instance [Caglio et al. \(2025\)](#).

Figure 10: COVID Deposit Growth



Notes: This figure shows the growth in various types of deposits over the COVID crisis. We index growth to January 2019. In Panel (b) we show the distribution of deposit growth over the entire COVID period for banks in our sample. We focus on Retail and Corporate deposits for ease of viewing.

transformation. The inflows of retail depositors are treated differently. Retail-oriented banks use these inflows to fund growth in real estate lending. This is consistent with our earlier findings. As above, we again find that retail-oriented banks respond to retail deposit inflows with an increase in cash holdings as well. This may reflect an operational lag: banks may need time to originate new loans or reallocate capital, and therefore temporarily park the new deposits.

We next assess how bank-level lending responses translate into real economic outcomes at the firm level. To do so, we merge loan-level supervisory data with firm-level financial data from Compustat. Our matching approach is as exhaustive as possible. We first match based on clear identifiers including ticker symbols, CUSIP and TIN. Where we cannot match firms, we use a combination of and firm name and headquarter zip code fuzzy matching. This linked dataset covers approximately 80% of the Compustat universe by firm assets. We calculate the degree to which a given borrower's lenders were affected by funding shocks. In cases where a borrower has multiple banks, we average across lenders. Our results look very similar if we take the largest lender or the maximum value for deposit changes (not reported for brevity). With this data, we are able to track how quasi exogenous changes in the funding environment of a firm's lender affect the borrowing firm's performance. Our sample is naturally limited to public firms and skews towards larger firms. We view these results as being indicative rather than causal.

Our dependent variables of interest are firm size and firm profitability. Our regressions are run at the firm*year level. We account for a firm size as well the industry in which it operates, which accounts for aggregate changes in demand industries may have faced during COVID. We ultimately want to

Table 11: Lending Changes on Deposit Changes during COVID

	(1) Δ C&I Lending	(2) Δ Securities	(3) Δ Cash/Reserves	(4) Δ Real Estate	(5) Δ C&I Lending	(6) Δ Securities	(7) Δ Cash/Reserves	(8) Δ Real Estate
Panel A								
Δ Corp. Dep.	-0.009 [0.043]	0.173*** [0.046]	0.510*** [0.179]	0.042 [0.039]	-0.017 [0.074]	0.127* [0.072]	0.050 [0.307]	-0.124** [0.061]
Share Corp. Dep.					-0.262 [0.271]	-0.181 [0.275]	1.255 [1.297]	0.143 [0.256]
Δ Corp. Dep. * Share Corp. Dep.					0.063* [0.031]	0.217 [0.304]	3.027* [1.704]	0.892*** [0.336]
Panel B								
Δ Retail Dep.	0.083 [0.075]	0.207*** [0.061]	0.713*** [0.263]	0.357*** [0.051]	0.149 [0.131]	-0.000 [0.102]	0.487 [0.431]	0.195** [0.077]
Share of Retail Deposits					-0.122 [0.284]	-0.220 [0.217]	-1.680* [0.904]	-0.494*** [0.162]
Δ Retail Dep. * Share of Retail Deposits					-0.314 [0.334]	0.690*** [0.258]	1.198 [1.048]	0.449** [0.188]
Panel C								
Δ NBFI Dep.	0.058* [0.035]	0.083*** [0.030]	0.212* [0.122]	0.016 [0.026]	0.036 [0.045]	0.067* [0.036]	0.036 [0.138]	-0.014 [0.029]
Share of NBFI Deposits					0.148 [0.280]	-0.214 [0.376]	-5.056*** [1.737]	-0.404 [0.366]
Δ NBFI Dep. * Share of NBFI Deposits					0.115 [0.208]	0.019 [0.197]	3.224*** [0.692]	0.174 [0.146]
N	190	190	190	190	190	190	190	190
Fixed Effects	Time				Time, Bank			

Notes: This table shows quarterly log changes in a bank's investments (the amount invested in C&I lending (including commitments), the value of securities held, the cash or reserves held, and real estate lending (including commercial and residential)) relative to contemporaneous changes in corporate, retail, or NBFI deposits during the COVID quarters (2020q2 to 2021q3). Columns (5) -(8) include bank fixed effects and interactions with the share of a bank's balance sheet the corresponding depositor-type represents. Standard errors are heteroskedasticity robust while *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

ascertain whether firms were able to grow – or increase their profitability – by investing in high NPV projects during a time of aggregate turmoil. Table 12 shows that corporate deposit inflows are either uncorrelated or negatively correlated with firm outcomes such as asset growth and profitability. In contrast, retail deposit inflows are weakly associated with increases in borrower assets and significantly associated with increases in profitability. This divergence suggests that the nature of the funding shock is meaningful for firm performance. We can see that retail deposits are associated with better loan terms, a fact that may translate to growth-related investment decisions where the loan's price is an important factor.

However, Table 13 once again makes clear that the nature of the funding shock *and* the nature of the bank matter. In columns (1) and (4) our independent variable of interest is growth in retail deposits, which we interact with the degree to which a bank's deposits are comprised of retail depositors. Columns (2) and (5) make use of corporate deposit changes and an interaction term for a bank's corporate deposit

Table 12: *Lending Changes on Deposit Changes during COVID*

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Δ Assets			EBITDA/ Assets				
Δ Retail Dep.	0.069 [0.204]			0.118 [0.214]	1.044*** [0.128]			1.025*** [0.130]
Δ Corp. Dep.		-0.162* [0.092]		-0.163 [0.108]		-0.155** [0.065]		-0.378*** [0.069]
Δ NBFI Dep.			-0.043 [0.068]	-0.010 [0.078]			0.203*** [0.039]	0.175 [0.165]
Fixed Effects	4-digit Industry, Year							
Controls	Firm Size							
R ²	0.089	0.089	0.089	0.089	0.16	0.15	0.16	0.17
N	6,559	6,559	6,559	6,559	6,448	6,448	6,448	6,448

Notes: This table shows the log changes in assets (columns (1)-(4) or firm profitability (columns (5) - (8)) in COMPUSTAT firms related to the change in the deposits of the lender associated with the given borrower. If a borrower has multiple lenders, we use the average change of all al borrower's lenders. We include controls for total firm borrowing/size as well as time and 4-digit industry fixed effects. Standard errors are clustered at the industry time level while *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

funding share. Columns (3) and (6) include both at the same time.

We can see that firms borrowing from retail deposit oriented banks are not as likely to grow their assets or their profitability. Conversely, corporate-depositor specialized banks convert corporate deposit inflows into new C&I lending, which allows their customers to grow profitably (columns (2) and (5)). When including both in the same regression, we can see that the interaction term of corporate deposit specialization and corporate deposit growth remains significant for borrower asset growth. Both terms remain significant when looking at profitability growth. NBFI-deposit oriented banks do not lend to C&I customers during COVID and their borrowers do not experience asset or profitability growth (not reported for brevity). These results corroborate what we saw earlier: banks with larger corporate deposit shares lower the cost of loans and increase loan sizes in response to corporate deposit inflows and visa-versa (though less pronounced) for retail deposit oriented banks, which are more focused on real estate.

Table 13: *Lending Changes on Deposit Changes during COVID*

	(1)	(2)	(3)	(4)	(5)	(6)
	Δ Assets		EBITDA/Assets			
Share Retail Deposits	0.114*** [0.030]		0.099** [0.050]	0.321*** [0.021]		0.001 [0.035]
Interaction: Δ Ret. * Share Ret. Dep	-1.200* [0.722]		-0.491 [0.846]	-6.624*** [0.490]		-3.734*** [0.568]
Δ Retail Dep.	0.550 [0.404]		0.277 [0.463]	4.033*** [0.279]		2.406*** [0.315]
Share Corp. Deposits		-0.132*** [0.039]	-0.030 [0.065]		-0.563*** [0.026]	-0.510*** [0.045]
Interaction: Δ Corp. * Share Corp. Dep		1.766** [0.691]	1.576** [0.739]		2.153*** [0.476]	1.291** [0.506]
Δ Corp. Dep.		-0.631*** [0.213]	-0.588** [0.231]		-0.639*** [0.148]	-0.368** [0.160]
Fixed Effects	4-digit Industry, Year					
Controls	Firm Size					
R ²	0.091	0.091	0.092	0.21	0.21	0.22
N	6,576	6,576	6,576	6,465	6,465	6,465

Notes: This table shows the log changes in assets (columns (1)-(4) or firm profitability (columns (5) - (8)) in COMPUSTAT firms related to the change in the deposits of the lender associated with the given borrower. If a borrower has multiple lenders, we use the average change of all al borrower's lenders. We interact our variables of interest with the share of a bank's balance sheet that is comprised of retail or corporate depositors. We again aggregate to the average of a firm's lenders if it has multiple. We include controls for total firm borrowing/size as well as time and 4-digit industry fixed effects. Standard errors are clustered at the industry time level while *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

6 Conclusion

This paper establishes a link between banks' depositor composition and their lending behavior, asset allocation, and response to liquidity shocks. Using granular supervisory data, we show that banks specialize not only in what they lend but also in whom they fund themselves with—and this depositor specialization is persistent and shapes bank business models in economically meaningful ways. Retail-funded banks, supported by more stable and cheaper deposits, offer larger and longer-term loans at lower rates and allocate more to real estate. Corporate- and NBFI-funded banks, facing more volatile funding, lend shorter, charge higher rates, and hold more liquid securities.

Funding shocks amplify these differences. Deposit inflows translate into distinct lending responses depending on the receiving bank's deposit specialization: retail banks channel new funds mainly to real estate or cash, whereas corporate banks expand C&I lending. Using the quasi-exogenous surge in

deposits during COVID-19, we show that these heterogeneous responses have real effects for borrowers, influencing firm growth and profitability.

Our findings imply that banks are not interchangeable conduits for credit or monetary policy. Asset and liability choices are jointly determined, and depositor composition is a defining dimension of bank heterogeneity. As a result, the impact of liquidity interventions—such as those during QE—depends critically on which banks receive the inflows. Recognizing this heterogeneity is key for theory, policy design, and prudential regulation.

It is worth noting that our findings also speak to a number of other long-running policy questions. One recent example may be the narrow bank discussions that have emerged. An NBFI-specialized bank in our data funds itself with the flightiest, most rate-sensitive deposits and holds almost exclusively reserves and Treasuries; it performs little maturity transformation and converts new funding into essentially no new credit. This is, in the limit, a narrow bank i.e. an institution that takes demandable deposits and backs them one-for-one with safe, liquid assets. The standard objection to such entities is that they would attract flighty wholesale funding away from credit-providing intermediaries without channeling it back to the real economy. However, it could be argued that NBFI-oriented banks already fill some part of that role without disintermediation concerns.

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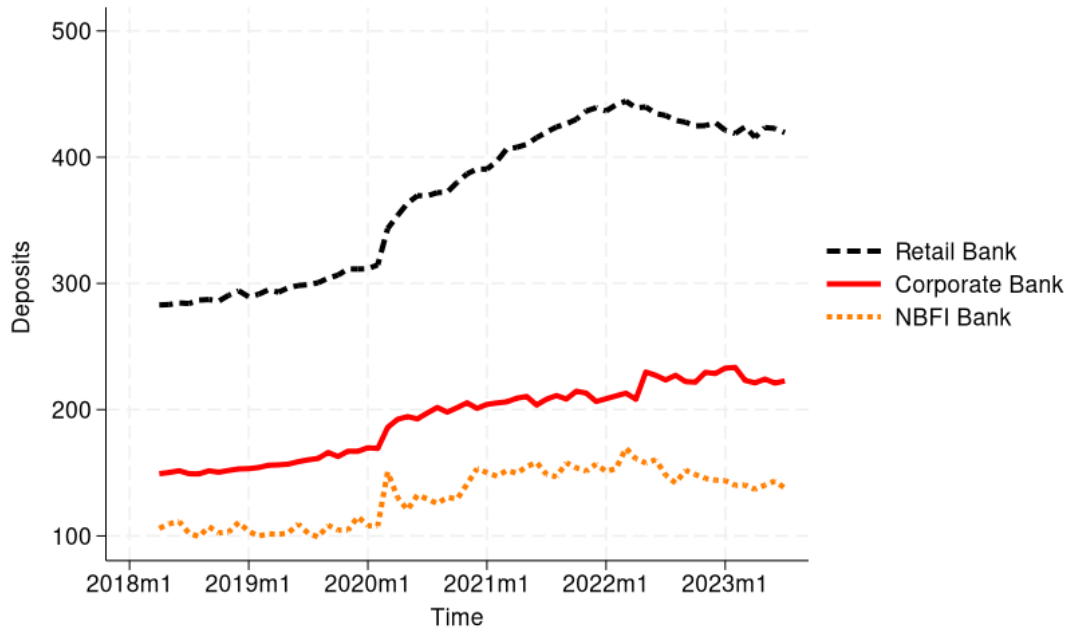
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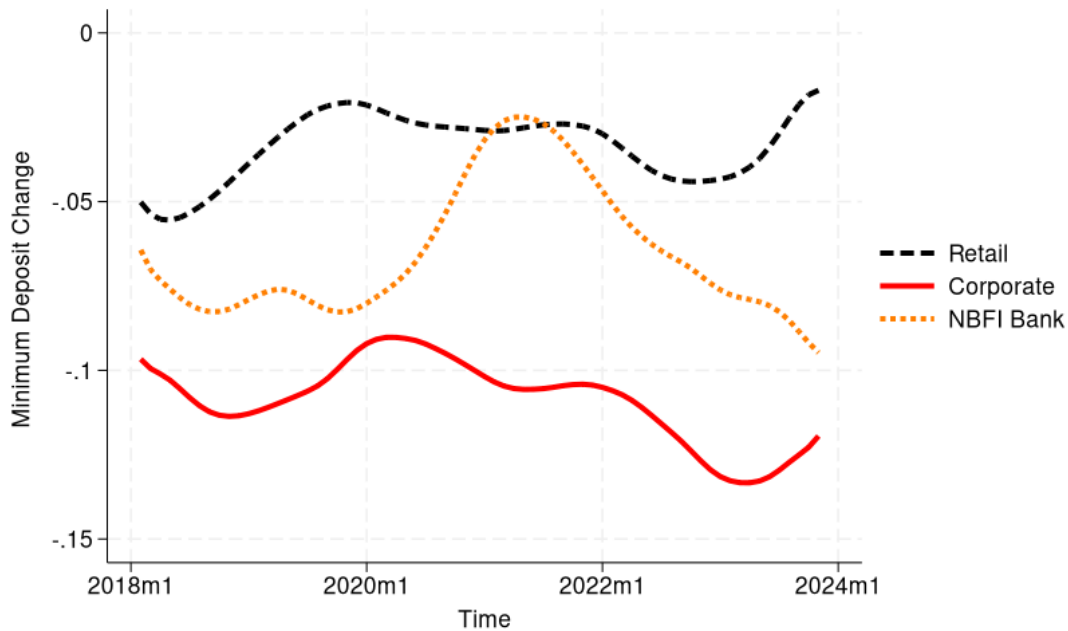
APPENDIX

Figure A.1: *Deposit Type Volume*



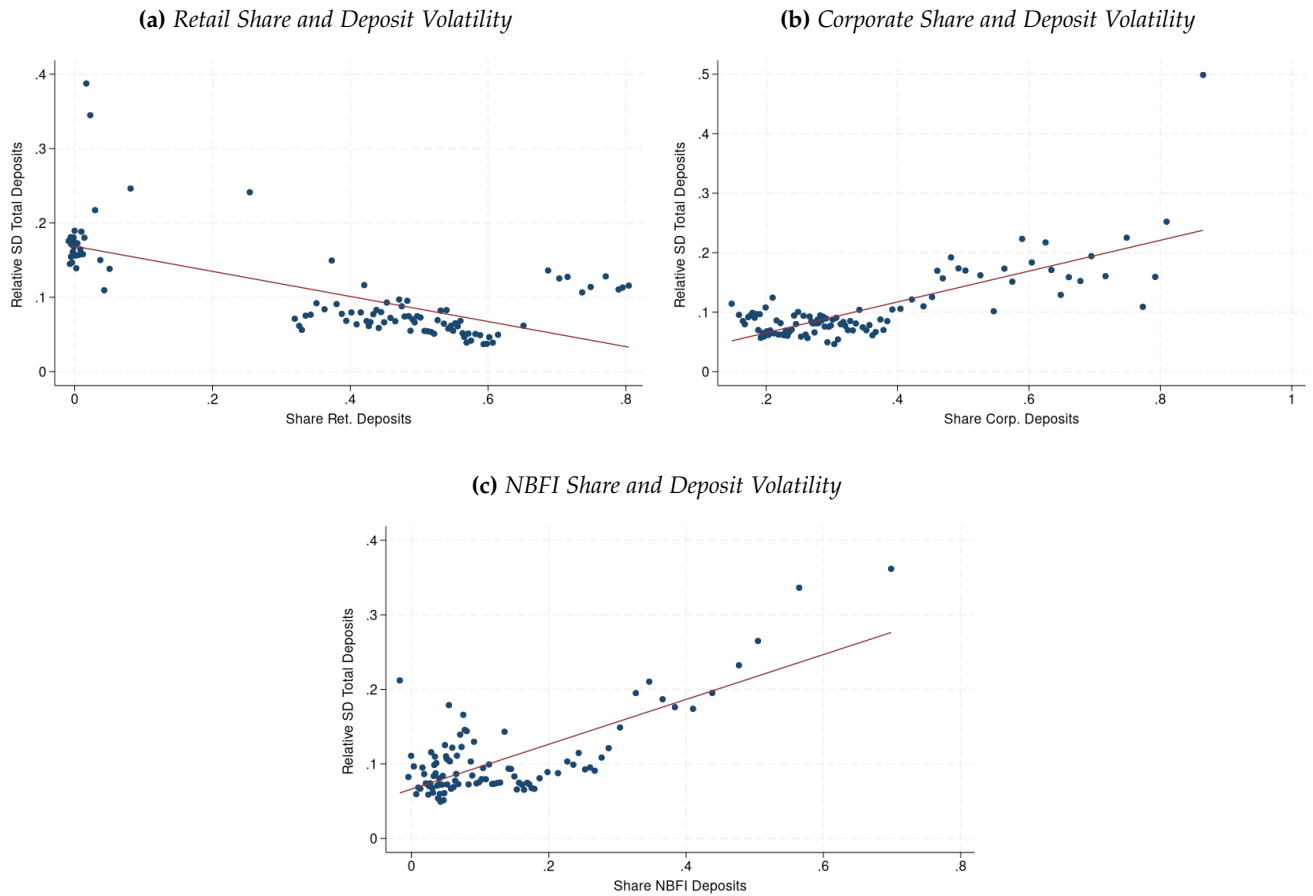
Notes: This figure splits banks in our sample by the largest depositor type (as in Figure 1) and shows the total volume of deposits by bank type.

Figure A.2: *Minimum Deposit Drop*



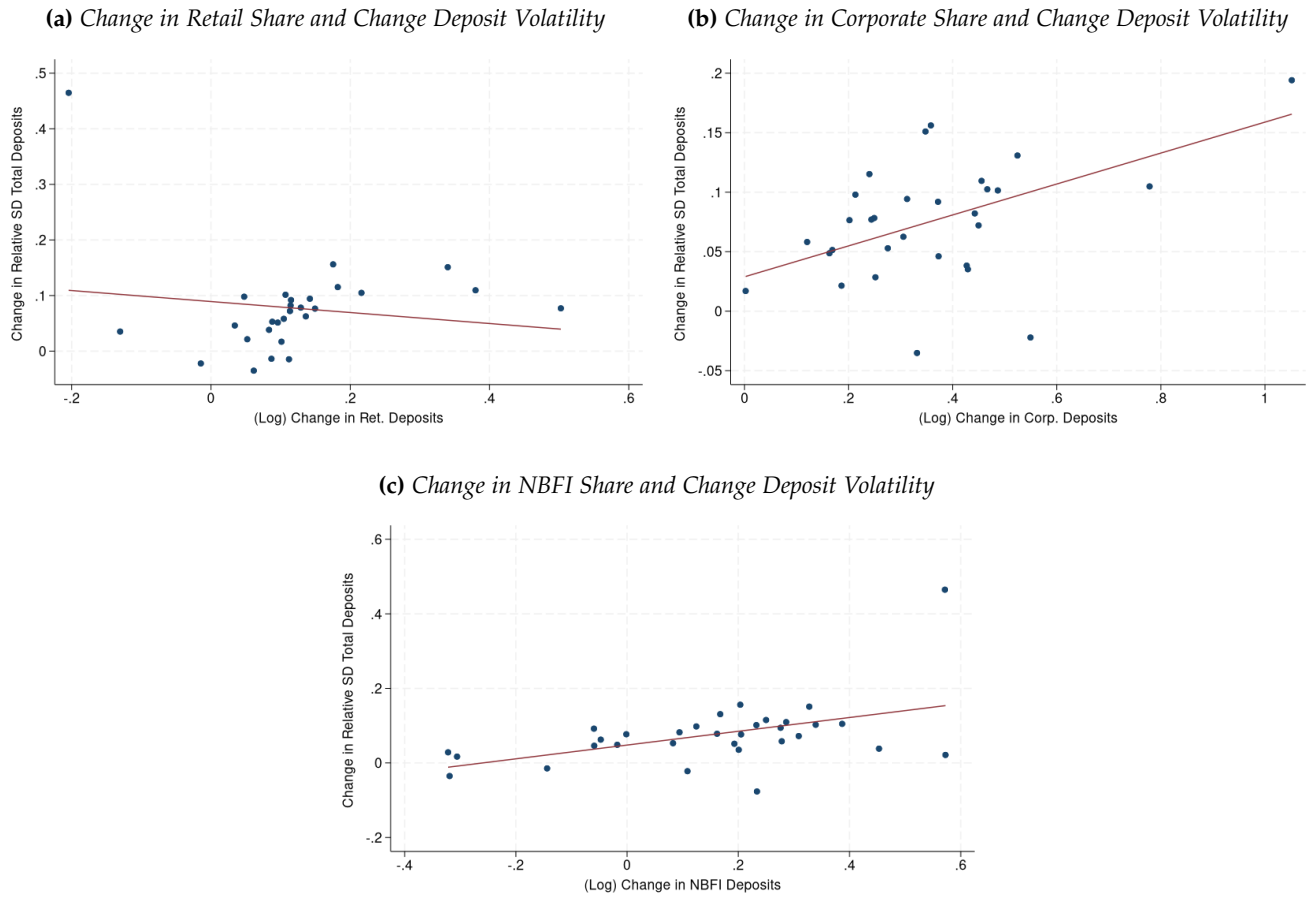
Notes: This figure shows the maximum drawdown any bank in our sample experienced from one quarter to the next. We split depositors into three types" Corporate, NBFI, or retail. We smooth our data for ease of viewing.

Figure A.3: Binscatter Deposit Types



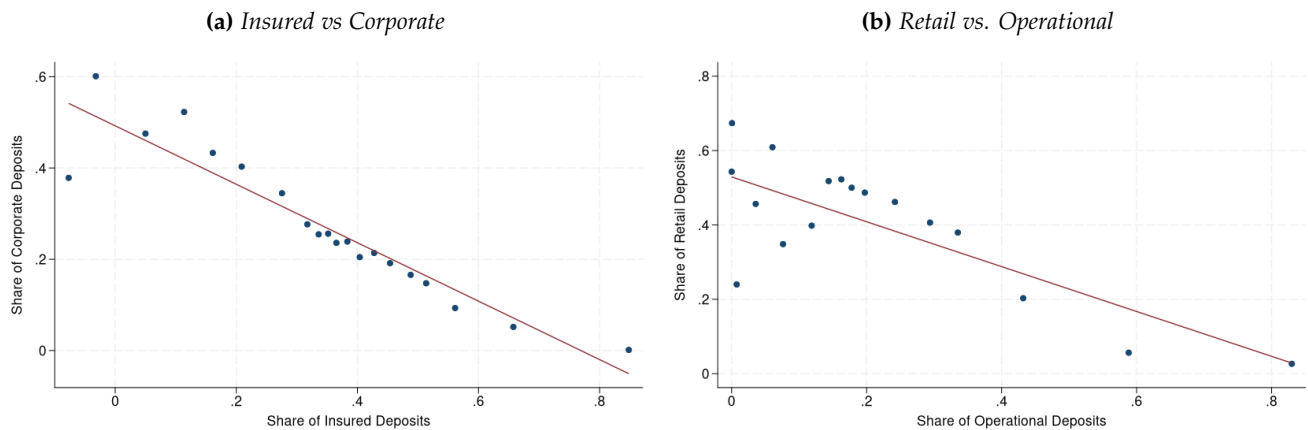
Notes: This figure shows binscatter plots relating the share of a bank's deposits originating from either NBFI, Corporate, or retail depositors to the relative standard deviation (12-month rolling standard deviation/ 12-month mean deposit base) of their total deposit base. We make use of bank*month data and each bin contains at least 5 observations.

Figure A.4: Binscatter Deposit Types (Changes)



Notes: Notes: This figure shows binscatter plots relating the change in the share of a bank's deposits originating from either NBFI, Corporate, or retail depositors to the change in relative standard deviation (12-month rolling standard deviation/12-month mean deposit base) of their total deposit base. We make use of bank*month data and each bin contains at least 5 observations.

Figure A.5: Binscatter Deposit Shares



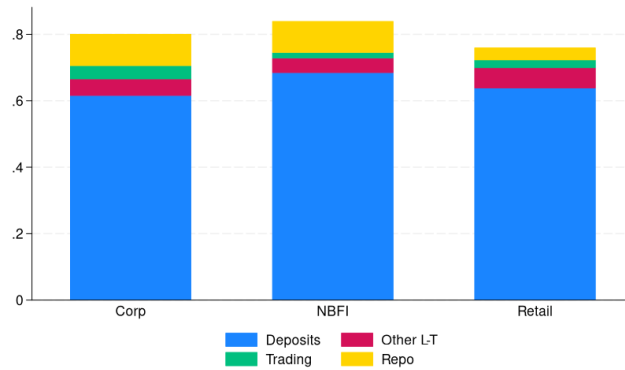
Notes: This figure shows binscatters that relate the (a) share of deposits that are insured relative to the share of deposits that are corporate (accounting for NBFI deposits) across bank-time observations in our sample or (b) the share that are retail deposits vs the share that are operational deposits across bank-time observations in our sample.

Table A.1: *Explanatory Power of Deposit Types*

	(1)	(2)	(3)	(4)	(5)	(6)
	Rel. Stdev. Deposits			Δ Deposits		
Share Corporate	0.065*** [0.013]	0.056*** [0.012]	0.046*** [0.007]	0.013 [0.012]	0.012 [0.012]	0.013 [0.012]
Share NBFI	0.048*** [0.014]	0.051*** [0.013]	0.036*** [0.007]	0.023* [0.013]	0.021* [0.013]	0.022* [0.013]
Share Retail	-0.039*** [0.014]	-0.045*** [0.013]	-0.011 [0.009]	-0.006 [0.013]	-0.004 [0.013]	-0.007** [0.003]
Share Insured	-0.022* [0.011]	-0.019* [0.010]	-0.013 [0.009]	0.016 [0.012]	0.015 [0.011]	0.015 [0.011]
Share Wholesale	0.051*** [0.004]	0.053*** [0.004]	0.045*** [0.003]	-0.000 [0.004]	-0.001 [0.004]	-0.000 [0.003]
Share Operational	0.000 [0.010]	-0.005 [0.010]	0.027*** [0.007]	-0.016 [0.010]	-0.013 [0.010]	-0.014 [0.011]
Log Total Deposits			-0.019*** [0.001]			0.001 [0.001]
Constant	0.052*** [0.009]	0.056*** [0.008]	0.263*** [0.009]	-0.001 [0.009]	-0.001 [0.008]	-0.012 [0.016]
Fixed Effects		Time	Time		Time	Time
R ²	.28	.4	.58	.0054	.089	.09
N	2664	2664	2664	2664	2664	2664

Notes: This Table shows the relationship between the share of a bank's deposits comprised of various depositor types and measures denoting deposit variability (either the relative standard deviation of deposits in columns (1) -(3) or the changes in deposits in columns (4)-(6). Standard errors are heteroskedasticity robust while *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Figure A.6: Liability Structure by Banktypes



Notes: This figure shows the balance sheet structure of different bank-types (differenced by depositor counter-parties that constitute the primary depositor type). We exclude foreign banks from this analysis, focusing on domestic banks.

Table A.2: Deposit Specialization on Relative Std. Dev. Deposits

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Rel. Stdev. Retail			Rel. Stdev. Corp.		Rel. Stdev. NBFI			
Retail Deposit Share	-0.286*** [0.026]	-0.288*** [0.026]	-0.448*** [0.122]						
[1em] Corporate Deposit Share				-0.178*** [0.053]	-0.197*** [0.050]	0.078 [0.094]			
NBFI Deposit Share							-0.210*** [0.021]	-0.214*** [0.021]	-0.153 [0.205]
Fixed Effects		Time	Bank, Time		Time	Bank, Time		Time	Bank, Time
Controls					Bank Size				
R ²	0.23	0.27	0.46	0.24	0.33	0.59	0.12	0.19	0.43
N	2,325	2,325	2,325	2,181	2,181	2,181	2,253	2,253	2,253

Notes: This Table shows the relationship between the share of deposits that make up the deposit base at a bank and the relative standard deviation of the deposit base. We include controls for bank size and fixed effects as indicated. Standard errors are heteroskedasticity robust while *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

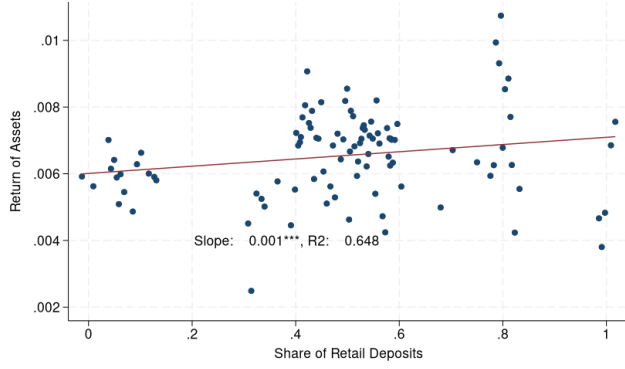
Table A.3: Deposit Specialization on Relative Std. Dev. Deposits – Above 9%

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Rel. Stdev. Retail			Rel. Stdev. Corp.		Rel. Stdev. NBFI			
Retail Deposit Share	-0.086*** [0.008]	-0.087*** [0.008]	-0.069 [0.059]						
Corporate Deposit Share				-0.021 [0.021]	-0.011 [0.019]	0.093 [0.078]			
NBFI Deposit Share							-0.068*** [0.011]	-0.069*** [0.011]	0.161*** [0.052]
Fixed Effects		Time	Bank, Time		Time	Bank, Time		Time	Bank, Time
Controls					Bank Size				
R ²	.25	.33	.67	.29	.43	.69	.19	.26	.79
N	2017	2017	2017	2362	2362	2361	1508	1508	1506

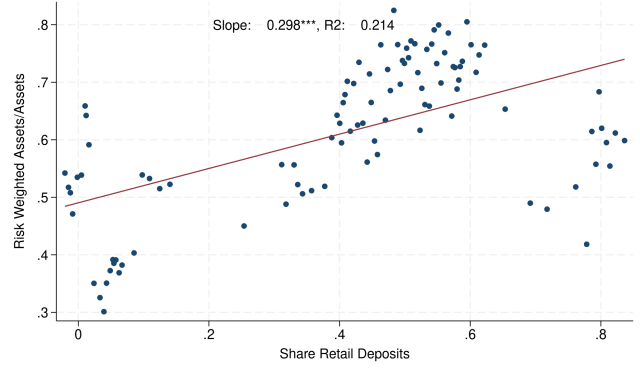
Notes: This Table shows the relationship between the share of deposits that make up the deposit base at a bank and the relative standard deviation of the deposit base. We include only bank-tome observations where a deposit counterparty make up more than 9 | 5 of total deposits. We include controls for bank size and fixed effects as indicated. Standard errors are heteroskedasticity robust while *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Figure A.7: Share of Deposits on Interest Income and Risk Weighted Assets

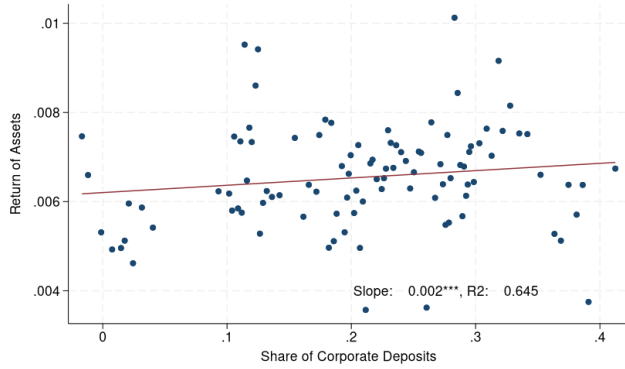
(a) Income vs. Retail Share



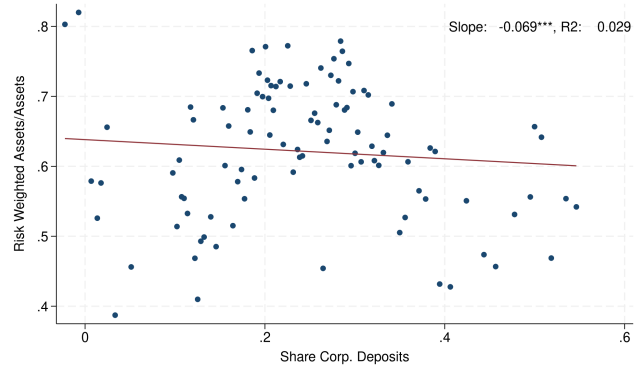
(b) RWA vs. Retail Share



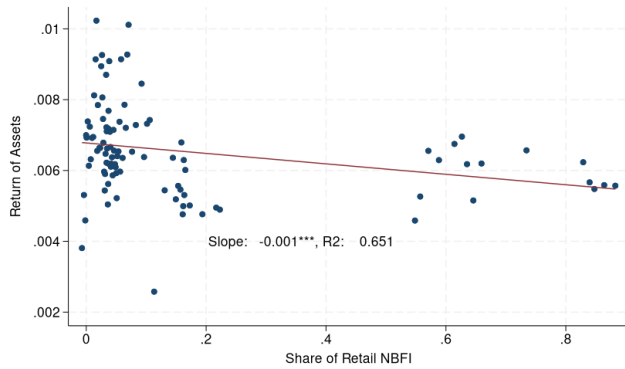
(c) Income vs. Corporate Share



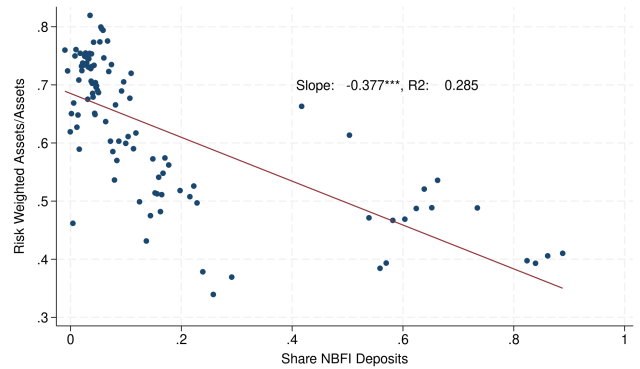
(d) RWA vs. Corporate Share



(e) Income vs. NBFI Share



(f) RWA vs. NBFI Share



This Figure shows the relationship between the share of retail, corporate, or NBFI depositors that comprise a bank's deposit base and the bank's income (panels (a)-(c)) or the bank's risk weighted assets (panels (d)-(f)). All points is binned collapsed bank-quarter level data with at least three bank-quarters per bin. We absorb time fixed effects. Each bin contains 5 bank-time observations.

Table A.4: RoA Shock on Deposit Changes

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Δ Retail	Δ Corp.	Δ NBFI	Δ Retail	Δ Corp.	Δ NBFI	Δ Retail	Δ Corp.	Δ NBFI
RoA Shock	-0.006 [0.013]	-0.032* [0.019]	-0.056* [0.031]	-0.006 [0.013]	-0.034* [0.020]	-0.057* [0.032]	-0.004 [0.013]	-0.037* [0.021]	-0.059* [0.033]
Tier-1 to Assets	-0.899 [0.568]	-1.203 [0.913]	-1.772 [1.416]	-0.901 [0.570]	-1.229 [0.912]	-1.781 [1.420]	-0.904 [0.570]	-1.211 [0.915]	-1.680 [1.418]
RWA to Assets	0.025 [0.090]	0.004 [0.140]	-0.414* [0.217]	0.026 [0.090]	0.031 [0.141]	-0.408* [0.219]	0.023 [0.090]	0.004 [0.141]	-0.401* [0.218]
Total Assets	0.000 [0.000]	0.000 [0.000]	-0.000 [0.000]	0.000 [0.000]	0.000 [0.000]	-0.000 [0.000]	0.000 [0.000]	0.000 [0.000]	-0.000 [0.000]
Interaction: Corporate Bank * RoA Shock				0.005 [0.034]	0.020 [0.052]	0.015 [0.081]			
Corporate Bank				-0.000 [0.040]	0.098 [0.062]	0.012 [0.096]			
Interaction: NBFI Bank * RoA Shock							-0.015 [0.027]	0.041 [0.042]	0.006 [0.065]
NBFI Bank							0.004 [0.031]	0.011 [0.048]	-0.100 [0.075]
Fixed Effects				Bank and Year*Quarter					
R ²	0.22	0.32	0.15	0.22	0.33	0.15	0.22	0.32	0.15
N	475	475	475	475	475	475	500	475	475

Notes: This Table shows the relationship between a drop in RoA that is larger than two standard deviations of a 2 year window (RoA shock) and depositor responses. Our dependent variable is log deposit changes for different depositor types. We account for bank capital and risk weighted assets as well as bank and time fixed effects. Standard errors are heteroskedasticity robust while *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Table A.5: Loan Risk and Deposit Shares

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	
	Rating	Non-Accrual	Rating	Non-Accrual	Rating	Non-Accrual	Rating	Non-Accrual	
Share Retail Dep.	0.864*** [0.005]	0.004*** [0.000]					0.293*** [0.014]	0.003*** [0.001]	
Share Corp. Dep.			-0.891*** [0.010]	0.002*** [0.001]			-0.840*** [0.016]	0.003*** [0.001]	
Share NBFI Dep.					-0.824*** [0.006]	-0.007*** [0.000]	-0.623*** [0.016]	-0.004*** [0.001]	
Fixed Effects	Collateral type, industry*year*quarter, loan purpose, loan type								
Controls	Total interest paid, bank assets, bank expenses, loan terms (other than the dep. var), loan rating								
R ²	0.26	0.32	0.28	0.26	0.32	0.28	0.26	0.32	0.28
N	3,339,792	3,339,792	3,339,792	3,339,792	3,339,792	3,339,792	3,339,792	3,339,792	3,339,792

Notes: This Table shows the relationship between loan ratings and loan performance and depositor compositions. Our dependent variable rating takes a value between 1-10, where larger numbers imply a lower rating. This is standardized across banks in our sample. our dependent variable non-accrual takes the value of 1 if a loan becomes non-accruing. Our independent variables are the share (between 0 and 10 of a bank's depositor based comprised of a certain type of depositor. We account for a host of loan characteristics, including loan rating where the dependent variable is whether a loan becomes non-accruing. Standard errors are heteroskedasticity robust while *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Table A.6: Balance Sheet Composition on Deposit Shares

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	C&I Lending Share	Securities Share	Cash/Reserves Share	Real Estate Share	C&I Lending Share	Securities Share	Cash/Reserves Share	Real Estate Share
Panel A								
Share of Retail Deposits	0.010 [0.015]	-0.180*** [0.011]	-0.147*** [0.012]	0.323*** [0.018]	0.040** [0.020]	0.029 [0.032]	-0.275*** [0.039]	0.201*** [0.038]
Panel B								
Share of Corp. Deposits	0.248*** [0.025]	0.037 [0.024]	-0.208*** [0.022]	-0.116*** [0.039]	-0.122*** [0.027]	0.176*** [0.043]	0.504*** [0.051]	-0.548*** [0.049]
Panel C								
Share of NBFI Deposits	-0.179*** [0.016]	0.198*** [0.013]	0.308*** [0.009]	-0.326*** [0.022]	0.036 [0.024]	-0.094** [0.039]	0.128*** [0.048]	-0.069 [0.047]
R2	.097	.54	.46	.51	.97	.92	.88	.96
N	652	652	652	652	652	652	652	652
Controls				Common Equity to Assets, RWA to Assets				
Fixed Effects	-	-	-	-			Bank	

Notes: This Table shows the relationship between a bank's asset-side composition (the share of a balance sheet invested in C&I lending (including commitments), value of securities held relative to all assets, the cash or reserves held relative to all assets and real estate lending (including commercial and residential) relative to all assets) and depositor composition (the share of depositors that are insured, the share that are short termist or demandable, the share that are corporations and the share that are retail). We include controls for bank equity and its risk weighted assets. Each cell represents an individual and independent regression. Standard errors are heteroskedasticity robust while *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

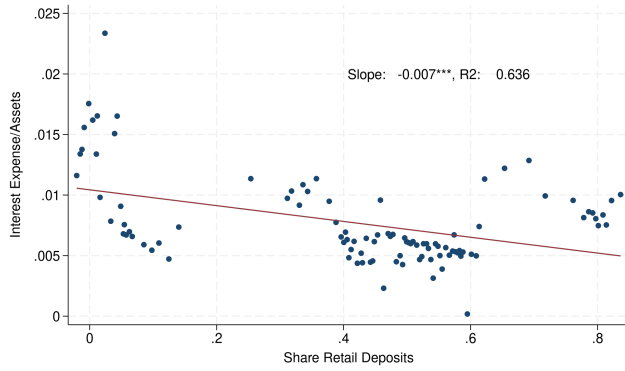
Table A.7: Lending Changes on Deposit Changes

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Δ C&I Lending	Δ Securities	Δ Cash/Reserves	Δ Real Estate	Δ C&I Lending	Δ Securities	Δ Cash/Reserves	Δ Real Estate
Panel A								
Δ Corp. Dep.	-0.001 [0.070]	0.182*** [0.047]	0.612*** [0.132]	0.254*** [0.046]	0.053 [0.120]	0.088 [0.080]	0.103 [0.223]	0.219*** [0.078]
Share Corp. Dep.					-0.050 [0.270]	-0.255 [0.181]	-0.222 [0.504]	0.033 [0.176]
Δ Corp. Dep. * Share Corp. Dep.					-0.318 [0.607]	0.671* [0.407]	3.218*** [1.133]	0.203 [0.395]
Panel B								
Δ Retail Dep.	0.235*** [0.043]	0.105*** [0.030]	0.095 [0.088]	0.104*** [0.030]	0.215*** [0.048]	-0.040 [0.030]	-0.023 [0.098]	-0.018 [0.031]
Share of Retail Deposits					0.016 [0.211]	0.022 [0.132]	-0.180 [0.426]	-0.521*** [0.134]
Δ Retail Dep. * Share of Retail Deposits					0.229 [0.252]	1.655*** [0.157]	1.367*** [0.507]	1.430*** [0.159]
Panel C								
Δ NBFI Dep.	0.071** [0.035]	0.091*** [0.024]	0.097 [0.069]	0.049** [0.024]	0.067* [0.039]	0.094*** [0.026]	0.003 [0.073]	0.067*** [0.026]
Share of NBFI Deposits					0.035 [0.319]	-0.157 [0.216]	-1.289** [0.604]	-0.696*** [0.212]
Δ NBFI Dep. * Share of NBFI Deposits					0.064 [0.254]	-0.002 [0.172]	1.902*** [0.482]	-0.120 [0.169]
N	475	475	475	475	475	475	475	475
Controls								
				Common Equity/Assets	RWA/Assets	Assets		Fixed Effects
				Time, Bank				

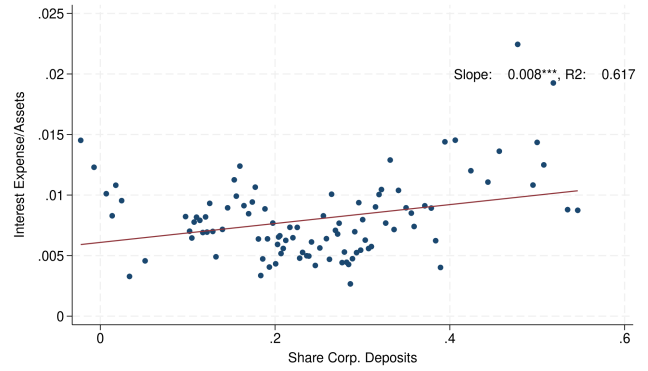
Notes: This table shows quarterly log changes in a bank's investments (the amount invested in C&I lending (including commitments), the value of securities held, the cash or reserves held, and real estate lending (including commercial and residential)) relative to contemporaneous changes in corporate, retail, or NBFI deposits. Columns (5) -(8) include bank fixed effects and interactions with the share of a bank's balance sheet the corresponding depositor-type represents. We include controls for bank equity and its risk weighted assets. Standard errors are heteroskedasticity robust while *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Figure A.8: Share of Deposits on Rate and Avg. Loan Rating

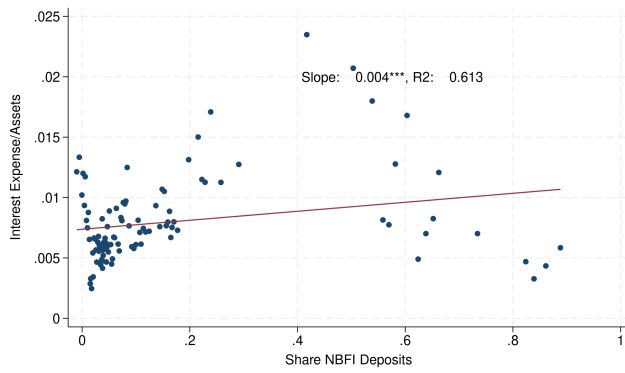
(a) Rate on Retail Share



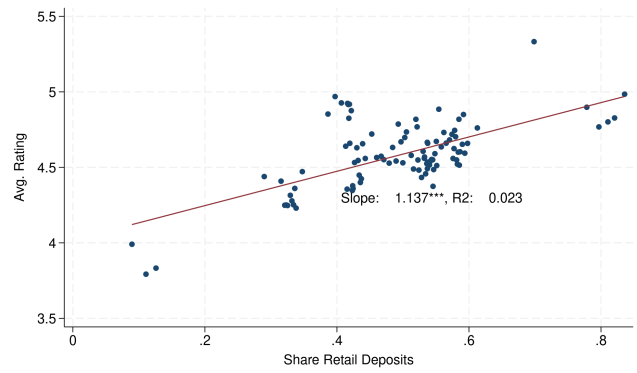
(b) Rate on Corporate Share



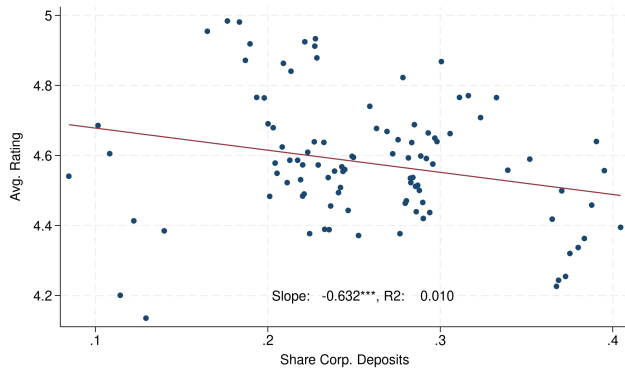
(c) Rate on NBFI Share



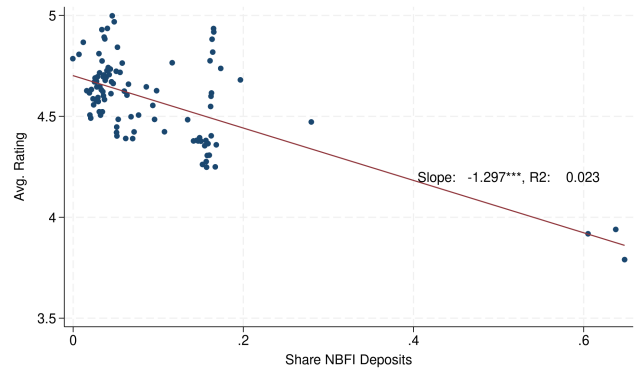
(d) Rating on Retail Share



(e) Rating on Corporate Share



(f) Rating on NBFI Share



This figure relates the share of a bank's deposits made up of a certain depositor type to the interest expenses to assets paid by the bank (Panels (a)-(c)) or the average rating received by loans of that bank (Panels (d)-(f)). Ratings are from 1-10 where larger ratings imply riskier loans.

Table A.8: *Securities Changes on Deposit Changes*

	(1) Δ Treas.	(2) Δ MBS	(3) Δ Other	(4) Δ CLO	(5) Δ Treas.	(6) Δ MBS	(7) Δ Other	(8) Δ CLO
Panel A								
Δ Corporate Deposits	0.057 [0.084]	0.026 [0.019]	0.101 [0.151]	0.412 [0.724]	0.050 [0.085]	0.029 [0.019]	0.100 [0.154]	0.463 [0.744]
Panel B								
Δ Retail Deposits	0.172 [0.121]	0.087*** [0.027]	0.084 [0.215]	-0.125 [0.484]	0.173 [0.125]	0.095*** [0.028]	0.097 [0.225]	-0.129 [0.507]
Panel C								
Δ NBFBI Deposits	0.277** [0.107]	0.100*** [0.024]	0.114 [0.212]	0.049 [0.426]	0.273** [0.109]	0.103*** [0.024]	0.087 [0.219]	0.036 [0.433]
N	500	500	500	500	500	500	500	500
Fixed Effects	-	-	-	-	Bank			

Notes: This Table relates changes to individual security types held by banks to changes in deposits. Each panel represents a separate set of regressions focused on an alternate independent variable. Columns (5) -(8) include bank fixed effects. Standard errors are heteroskedasticity robust while *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.